Supplementary Material

Molecular Dynamics Simulation of Thiol-ene 'click' Copolymerized and Blended Carboxymethyl Cellulose and 4-arm Polyethylene Glycol Used as Ciprofloxacin Drug Carrier

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aqueous

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Table S1. System configurations for aqueous media

System Name	Initial box dimension (nm×nm×nm)	Final box dimension (nm×nm×nm)	Number of polymer atoms	Number drugs atoms	Number of water atoms	Total number of atoms
C_NCMC_TPEG	10.81×10.81×10.81	8.76×8.76×8.76	7488	84	80000	87572
C_NCMC_SPEG	11.74×11.74×11.74	8.75×8.75×8.75	7024	84	80000	87108
C_TCMC_NPEG	12.15×12.15×12.15	8.77×8.77×8.77	7488	84	80000	87572
C_TCMC_SPEG	12.10×12.10×12.10	8.75×8.75×8.75	7024	84	80000	87108
B_NCMC_TPEG	12.48×12.48×12.48	8.76×8.76×8.76	7488	84	80000	87572
B_NCMC_SPEG	12.59×12.59×12.59	8.74×8.74×8.74	7024	84	80000	87108
B_TCMC_NPEG	12.03×12.03×12.03	8.77×8.77×8.77	7488	84	80000	87572
B_TCMC_SPEG	12.82×12.82×12.82	8.75×8.75×.75	7120	84	80000	87204

Table S2. System configurations for PBS media

System Name	Initial box dimension (nm×nm×nm)	Final box dimension (nm×nm×nm)	Number of polymer atoms	Number drugs atoms	Number of water atoms	Number of ions atom	Total number of atoms
C_NCMC_TPEG	20.00×20.00×20.00	8.77×8.77×8.77	7488	84	80000	296	87868
C_NCMC_SPEG	20.00×20.00×20.00	8.76×8.76×8.76	7024	84	80000	296	87404
C_TCMC_NPEG	20.00×20.00×20.00	8.78×8.78×8.78	7488	84	80000	296	87868
C_TCMC_SPEG	20.00×20.00×20.00	8.76×8.76×8.76	7024	84	80000	296	87404
B_NCMC_TPEG	20.00×20.00×20.00	8.78×8.78×8.78	7488	84	80000	296	87868
B_NCMC_SPEG	20.00×20.00×20.00	8.75×8.75×8.75	7024	84	80000	296	87404
B_TCMC_NPEG	20.00×20.00×20.00	8.78×8.78×8.78	7488	84	80000	296	87868
B_TCMC_SPEG	20.00×20.00×20.00	8.76×8.76×8.76	7120	84	80000	296	87500

Table S3. Average number of H-bonds

		NCMC_TPEG	NCMC_SPEG	TCMC_NPEG	SCMC_NPEG
blend_pbs	drug_pbs	2	2	1	1
	poly_drug	1	1	2	1
	poly_pbs	316	323	344	332
	poly_poly	173	165	167	171
	poly1_drug	1	0	2	1
	poly1_sol	208	208	217	196
	poly1_poly2	19	15	14	15
	poly2_drug	0	0	0	0
	poly2_sol	108	115	126	136
blend_water	drug_sol	2	2	1	2
	poly_drug	1	1	1	1
	poly_sol	338	334	342	329
	poly_poly	182	165	169	170
	poly1_drug	1	1	1	1
	poly1_sol	176	197	217	197
	poly1_poly2	21	13	12	17
	poly2_drug	0	0	0	0
	poly2_sol	162	137	125	132
copolymer_pbs	drug_sol	1	2	2	2
	poly_drug	1	1	1	1
	poly_sol	316	306	332	294
	poly_poly	173	157	162	156
	poly1_drug	1	1	1	1
	poly1_sol	193	196	225	188
	poly1_poly2	36	34	30	25
	poly2_drug	0	0	0	0

	poly2_sol	124	110	108	106
	drug_sol	1	2	1	2
	poly_drug	1	1	1	1
	poly_sol	341	325	338	319
	poly_poly	172	153	163	158
copolymer_water	poly1_drug	1	1	1	1
	poly1_sol	203	200	223	205
	poly1_poly2	36	30	30	26
	poly2_drug	0	0	0	0
	poly2_sol	138	125	115	113

Table S4. The radial distribution functions and coordination number of the systems

component name	System name	solvation shell(Å)	*CN*	solvation shell(Å)	*CN*
oxygen of the hydroxyl in the ciprofloxacin to hydrogen of the water	B_NCMC_TPEG_PBS	2.55	0.48	-	-
	B_NCMC_SPEG_PBS		0.38		-
	B_TCMC_NPEG_PBS		0.50		-
	B_SCMC_NPEG_PBS		0.05		-
	B_NCMC_TPEG_HOH	2.45	0.37	-	-
	B_NCMC_SPEG_HOH		0.36		-
	B_TCMC_NPEG_HOH		0.20		-
	B_SCMC_NPEG_HOH		0.55		-
	C_NCMC_TPEG_PBS	2.55	0.00	-	-
	C_NCMC_SPEG_PBS		0.63		-
	C_TCMC_NPEG_PBS		0.42		-
	C_SCMC_NPEG_PBS		0.34		-
	C_NCMC_TPEG_HOH	2.45	0.10	-	-
	C_NCMC_SPEG_HOH		0.32		-
	C_TCMC_NPEG_HOH		0.65		-
	C_SCMC_NPEG_HOH		0.33		-
Nitrogen of amine of Ciprofloxacin to the unrreacted ene	B_NCMC_TPEG_PBS	8.25	0.04	-	-
	B_NCMC_SPEG_PBS	6.05	0.03		-
	B_TCMC_NPEG_PBS	7.15	0.04		-
	B_SCMC_NPEG_PBS	10.05	0.54		-
	B_NCMC_TPEG_HOH	5.45	0.01	-	-
	B_NCMC_SPEG_HOH	10.05	0.20		-
	B_TCMC_NPEG_HOH	8.25	0.08		-
	B_SCMC_NPEG_HOH	8.25	0.02		-
	C_NCMC_TPEG_PBS	7.25	0.08	-	-
	C_NCMC_SPEG_PBS	8.25	0.06		-

	C_TCMC_NPEG_PBS	13.85	0.10		-
	C_SCMC_NPEG_PBS	8.25	0.07		-
	C_NCMC_TPEG_HOH	8.65	0.19		-
	C_NCMC_SPEG_HOH	14.15	2.18		-
	C_TCMC_NPEG_HOH	14.55	0.13	-	-
	C_SCMC_NPEG_HOH	9.65	0.08		-
Fluorine of Ciprofloxacin to the unreacted ene	B_NCMC_TPEG_PBS	8.85	0.12	-	-
	B_NCMC_SPEG_PBS	9.85	0.12	-	-
	B_TCMC_NPEG_PBS	13.05	0.03	-	-
	B_SCMC_NPEG_PBS	10.25	0.41	-	-
	B_NCMC_TPEG_HOH	10.45	0.13	-	-
	B_NCMC_SPEG_HOH	8.95	0.05	-	-
	B_TCMC_NPEG_HOH	8.65	0.08	-	-
	B_SCMC_NPEG_HOH	6.95	0.02	-	-
	C_NCMC_TPEG_PBS	5.65	0.04	-	-
	C_NCMC_SPEG_PBS	9.25	0.16	-	-
	C_TCMC_NPEG_PBS	7.45	0.02	-	-
	C_SCMC_NPEG_PBS	6.75	0.07	-	-
	C_NCMC_TPEG_HOH	6.25	0.01	-	-
	C_NCMC_SPEG_HOH	10.75	1.76	-	-
	C_TCMC_NPEG_HOH	9.75	0.04	-	-
	C_SCMC_NPEG_HOH	8.15	0.07	-	-
oxygen of the hydroxyl in the ciprofloxacin to unreacted ene	B_NCMC_TPEG_PBS	5.85	0.03	-	-
	B_NCMC_SPEG_PBS	5.85	0.02	-	-
	B_TCMC_NPEG_PBS	4.35	0.00	-	-
	B_SCMC_NPEG_PBS	3.95	0.01	-	-
	B_NCMC_TPEG_HOH	8.95	0.14	-	-
	B_NCMC_SPEG_HOH	8.45	0.12	-	-
	B_TCMC_NPEG_HOH	5.45	0.02	-	-

	B_SCMC_NPEG_HOH	9.35	0.03	-	-
	C_NCMC_TPEG_PBS	8.75	0.22	-	-
	C_NCMC_SPEG_PBS	10.05	0.14	-	-
	C_TCMC_NPEG_PBS	6.45	0.00	-	-
	C_SCMC_NPEG_PBS	9.15	0.10	-	-
	C_NCMC_TPEG_HOH	10.45	0.20	-	-
	C_NCMC_SPEG_HOH	11.65	1.95	-	-
	C_TCMC_NPEG_HOH	12.15	0.02	-	-
	C_SCMC_NPEG_HOH	11.85	0.15	-	-
Nitrogen of the ciprofloxacin to unreacted ene	B_NCMC_TPEG_PBS	9.75	0.10	-	-
	B_NCMC_SPEG_PBS	10.25	0.14	-	-
	B_TCMC_NPEG_PBS	7.25	0.03	-	-
	B_SCMC_NPEG_PBS	5.95	0.07	-	-
	B_NCMC_TPEG_HOH	8.05	0.13	-	-
	B_NCMC_SPEG_HOH	9.65	0.18	-	-
	B_TCMC_NPEG_HOH	7.15	0.05	-	-
	B_SCMC_NPEG_HOH	10.25	0.04	-	-
	C_NCMC_TPEG_PBS	10.05	0.22	-	-
	C_NCMC_SPEG_PBS	9.55	0.14	-	-
	C_TCMC_NPEG_PBS	9.95	0.04	-	-
	C_SCMC_NPEG_PBS	8.85	0.08	-	-
	C_NCMC_TPEG_HOH	9.05	0.20	-	-
	C_NCMC_SPEG_HOH	11.55	1.75	-	-
	C_TCMC_NPEG_HOH	10.75	0.01	-	-
	C_SCMC_NPEG_HOH	7.45	0.05	-	-
Ether of PEG to the carboxylic acid of CMC	B_NCMC_TPEG_PBS	3.65	0.05	6.75	0.68
	B_NCMC_SPEG_PBS		0.13		1.49
	B_TCMC_NPEG_PBS		0.04		0.56
	B_SCMC_NPEG_PBS		0.16		1.82
	B_NCMC_TPEG_HOH	3.75	0.06	6.45	0.55

	B_NCMC_SPEG_HOH		0.08		0.81
	B_TCMC_NPEG_HOH		0.04		0.40
	B_SCMC_NPEG_HOH		0.10		0.88
	C_NCMC_TPEG_PBS	3.65	0.09	6.65	1.13
	C_NCMC_SPEG_PBS		0.14		1.53
	C_TCMC_NPEG_PBS		0.12		1.28
	C_SCMC_NPEG_PBS		0.09		1.20
	C_NCMC_TPEG_HOH	3.65	0.08	6.65	1.09
	C_NCMC_SPEG_HOH		0.12		1.48
	C_TCMC_NPEG_HOH		0.10		1.22
	C_SCMC_NPEG_HOH		0.09		1.15
Ether of PEG to the hydroxyl of CMC	B_NCMC_TPEG_PBS	3.45	0.07	6.35	1.47
	B_NCMC_SPEG_PBS		0.06		1.07
	B_TCMC_NPEG_PBS		0.03		0.46
	B_SCMC_NPEG_PBS		0.05		1.04
	B_NCMC_TPEG_HOH	3.45	0.06	5.55	0.96
	B_NCMC_SPEG_HOH		0.06		0.92
	B_TCMC_NPEG_HOH		0.05		0.65
	B_SCMC_NPEG_HOH		0.06		0.86
	C_NCMC_TPEG_PBS	3.25	0.08	5.45	1.37
	C_NCMC_SPEG_PBS	3.25	0.11	5.45	1.92
	C_TCMC_NPEG_PBS	4.05	0.30	10.85	8.55
	C_SCMC_NPEG_PBS	3.25	0.08	5.45	1.61
	C_NCMC_TPEG_HOH	3.45	0.12	5.45	1.31
	C_NCMC_SPEG_HOH		0.16		1.76
	C_TCMC_NPEG_HOH		0.12		1.51
	C_SCMC_NPEG_HOH		0.13		1.63
unreacted the sulphur to the unreacted ene	B_NCMC_TPEG_PBS	5.65	0.10	-	-
	B_NCMC_SPEG_PBS		0.09	-	-
	B_TCMC_NPEG_PBS		0.12	-	-

	B_SCMC_NPEG_PBS		0.05	-	-
	B_NCMC_TPEG_HOH	4.25	0.03	-	-
	B_NCMC_SPEG_HOH		0.03	-	-
	B_TCMC_NPEG_HOH		0.01	-	-
	B_SCMC_NPEG_HOH		0.02	-	-
	C_NCMC_TPEG_PBS	6.95	0.01	-	-
	C_NCMC_SPEG_PBS	6.25	0.06	-	-
	C_TCMC_NPEG_PBS	7.15	0.11	-	-
	C_SCMC_NPEG_PBS	11.55	0.70	-	-
	C_NCMC_TPEG_HOH	6.95	0.03	-	-
	C_NCMC_SPEG_HOH	10.75	1.17	-	-
	C_TCMC_NPEG_HOH	4.95	0.02	-	-
	C_SCMC_NPEG_HOH	7.95	0.29	-	-

Table S5. MSD of blend composites

component name	System name	Time (ns)	β	Time (ns)	β	Time (ns)	β	Time (ns)	β
Oxygen of Hydroxyl of Ciprofloxacin	B_NCMC_TPEG_HOH	0-22.3	1.00	22.3-30.0	0.12	-	-	-	-
	B_NCMC_SPEG_HOH	0-12.2	1.00	12.2-24.1	-0.37	24.1-30.0	0.97	-	-
	B_TCMC_NPEG_HOH	0-7.8	0.89	7.8-23.0	0.63	23.0-30.0	1.00	-	-
	B_SCMC_NPEG_HOH	0-9.2	0.70	9.2-22.6	-0.46	22.6-30.0	3.91	-	-
	B_NCMC_TPEG_PBS	0-7.4	0.79	7.4-14.9	0.21	14.9-22.3	0.85	22.3-30.0	-0.96
	B_NCMC_SPEG_PBS	0-15.8	1.00	15.8-30.0	1.00	-	-	-	-
	B_TCMC_NPEG_PBS	0-18.3	0.66	18.3-24.3	-0.85	24.3-30	2.16	-	-
	B_SCMC_NPEG_PBS	0-22.3	0.66	22.4-30.0	1.96	-	-	-	-
Unreacted ene group in blend composites	B_NCMC_TPEG_HOH	0-18.2	0.91	18.2-24.1	-0.80	24.1-30.0	1.27	-	-
	B_NCMC_SPEG_HOH	0-13.1	0.92	13.1-22.5	-0.08	22.5-30.0	-0.59	-	-
	B_TCMC_NPEG_HOH	0-12.1	0.85	12.1-24.1	0.50	24.1-30.0	1.67	-	-
	B_SCMC_NPEG_HOH	0-9.2	0.59	9.2-22.6	0.06	22.6-30.0	4.23	-	-
	B_NCMC_TPEG_PBS	0-7.5	0.79	7.5-22.3	0.80	22.3-30.0	0.51	-	-

	B_NCMC_SPEG_PBS	0-7.8	0.80	7.8-22.5	0.29	22.5-30.0	2.42	-	-
	B_TCMC_NPEG_PBS	0-22.6	0.66	22.6-30.0	1.45	-	-	-	-
	B_SCMC_NPEG_PBS	0-22.1	0.76	22.1-30.0	2.11	-	-	-	-
Unreacted thiol group in blend composites	B_NCMC_TPEG_HOH	0-19.8	0.85	19.8-23.7	-1.08	23.7-30.0	0.83	-	-
	B_NCMC_SPEG_HOH	0-14.2	0.92	14.2-22.5	-0.48	22.5-30.0	-0.45	-	-
	B_TCMC_NPEG_HOH	0-15.2	0.79	15.2-21.8	0.45	21.8-30.0	1.67	-	-
	B_SCMC_NPEG_HOH	0-9.2	0.57	9.2-22.6	0.26	22.6-30.0	3.34	-	-
	B_NCMC_TPEG_PBS	0-7.3	0.74	7.3-22.5	0.57	22.5-30.0	1.82	-	-
	B_NCMC_SPEG_PBS	0-7.8	0.75	7.8-22.5	0.55	22.5-30.0	1.60	-	-
	B_TCMC_NPEG_PBS	0-18.2	0.72	18.2-23.5	0.04	23.5-30.0	0.70	-	-
	B_SCMC_NPEG_PBS	0-22.1	0.67	22.1-30.0	1.56	-	-	-	-

Table S6. MSD of copolymer composites

component name	System name	Time (ns)	β	Time (ns)	β	Time (ns)	β	Time (ns)	β
Oxygen of Hydroxyl of Ciprofloxacin	C_NCMC_TPEG_HOH	0-14.2	0.61	14.2-22.5	0.28	22.5-30.0	3.78	-	-
	C_NCMC_SPEG_HOH	0-23.3	1.00	23.3-30.0	-0.40	-	-	-	-
	C_TCMC_NPEG_HOH	0-7.6	0.79	7.6-22.6	0.11	22.6-30.0	2.43	-	-
	C_SCMC_NPEG_HOH	0-15.1	0.84	15.1-30.0	0.35	-	-	-	-
	C_NCMC_TPEG_PBS	0-30.0	1.00	-	-	-	-	-	-
	C_NCMC_SPEG_PBS	0-22.1	0.82	22.1-30.0	0.69	-	-	-	-
	C_TCMC_NPEG_PBS	0-22.5	0.92	22.5-30.0	-0.13	-	-	-	-
	C_SCMC_NPEG_PBS	0-22.6	1.11	22.6-30.0	0.01	-	-	-	-
Unreacted ene group in copolymer composites	C_NCMC_TPEG_HOH	0-10.5	0.71	10.5-22.5	0.12	22.5-30.0	1.88	-	-
	C_NCMC_SPEG_HOH	0-23.1	0.97	23.1-30.0	0.19	-	-	-	-
	C_TCMC_NPEG_HOH	0-7.5	0.75	7.5-22.7	0.13	22.7-30.0	2.82	-	-
	C_SCMC_NPEG_HOH	0-15.2	0.97	15.2-30.0	0.29	-	-	-	-
	C_NCMC_TPEG_PBS	0-22.3	1.01	22.3-30.0	0.09	-	-	-	-
	C_NCMC_SPEG_PBS	0-30.0	0.91	-	-	-	-	-	-

	C_TCMC_NPEG_PBS	0-22.3	1.23	22.3-30.0	0.09	-	-	-	-
	C_SCMC_NPEG_PBS	0-22.6	1.14	22.6-30.0	0.01	-	-	-	-
Unreacted thiol group in copolymer composites	C_NCMC_TPEG_HOH	0-10.5	0.68	10.5-22.5	0.30	22.5-30.0	2.16	-	-
	C_NCMC_SPEG_HOH	0-23.1	0.93	23.1-30.0	0.03	-	-	-	-
	C_TCMC_NPEG_HOH	0-7.5	0.79	7.5-22.7	-0.04	22.7-30.0	2.91	-	-
	C_SCMC_NPEG_HOH	0-15.2	0.97	15.2-30.0	0.31	-	-	-	-
	C_NCMC_TPEG_PBS	0-22.3	0.82	22.3-30.0	-0.39	-	-	-	-
	C_NCMC_SPEG_PBS	0-30.0	0.88	-	-	-	-	-	-
	C_TCMC_NPEG_PBS	0-22.3	1.24	22.3-30.0	0.10	-	-	-	-
	C_SCMC_NPEG_PBS	0-22.6	1.11	22.6-30.0	-0.46	-	-	-	-

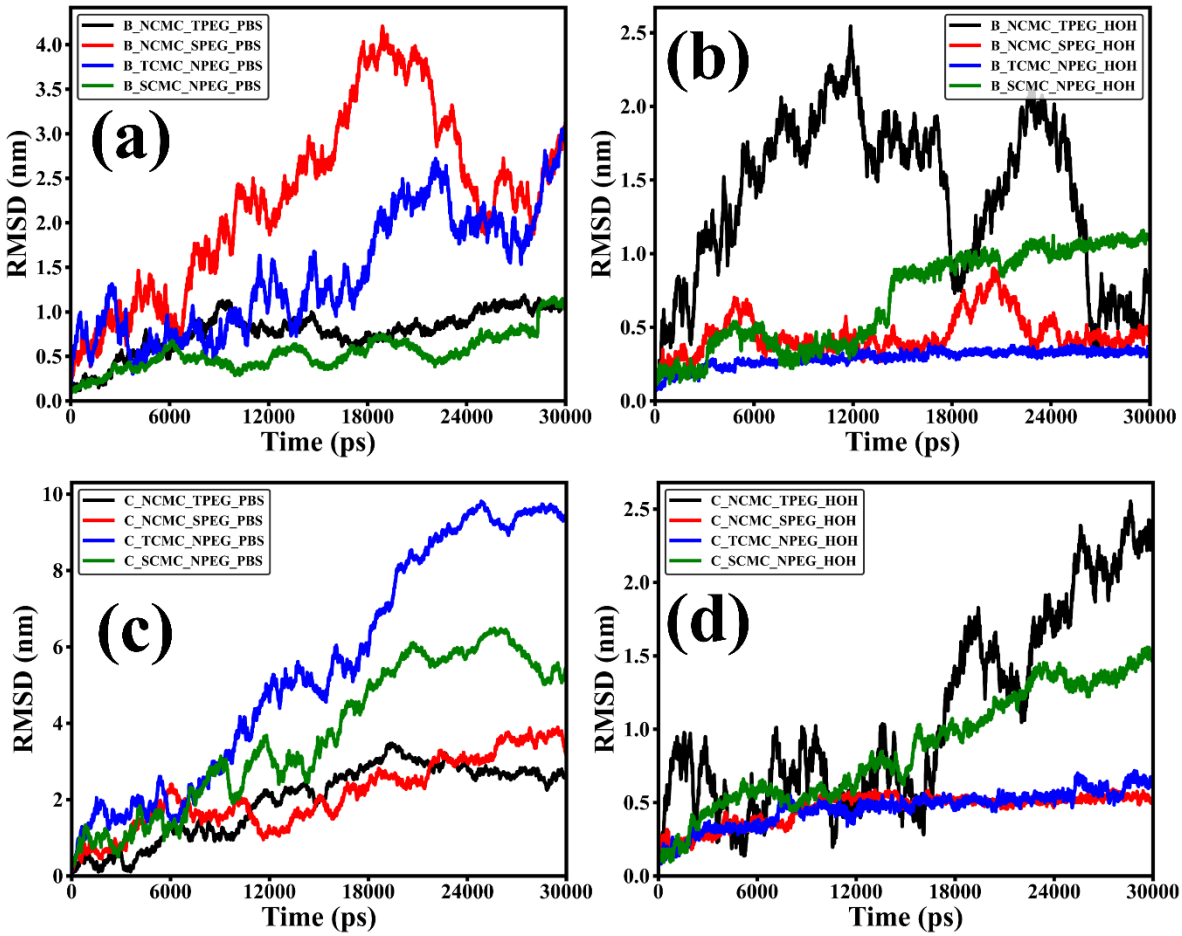


Figure S1. RMSD of drug-CMC based polymer framework to drug-CMC based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

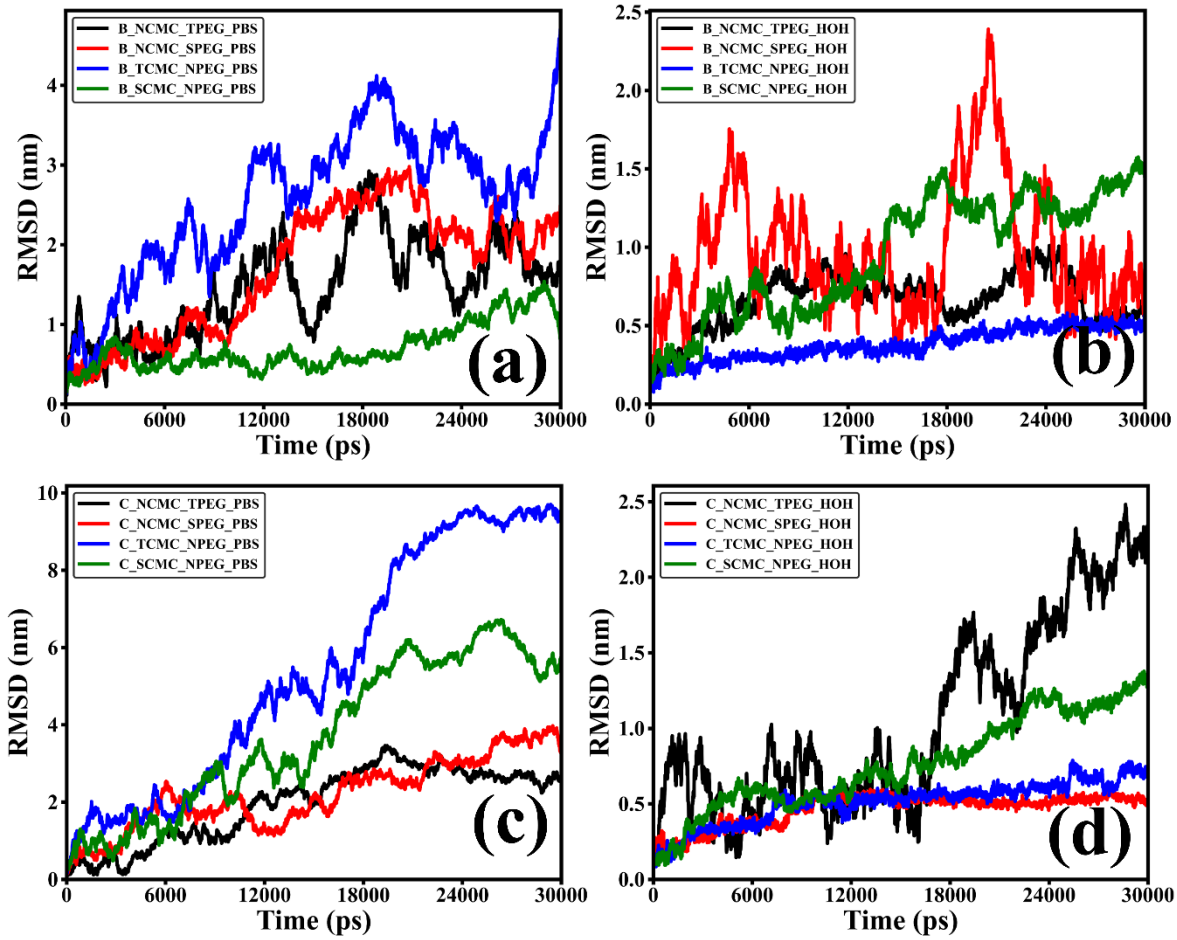


Figure S2. RMSD of drug-PEG based polymer framework to drug-PEG based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

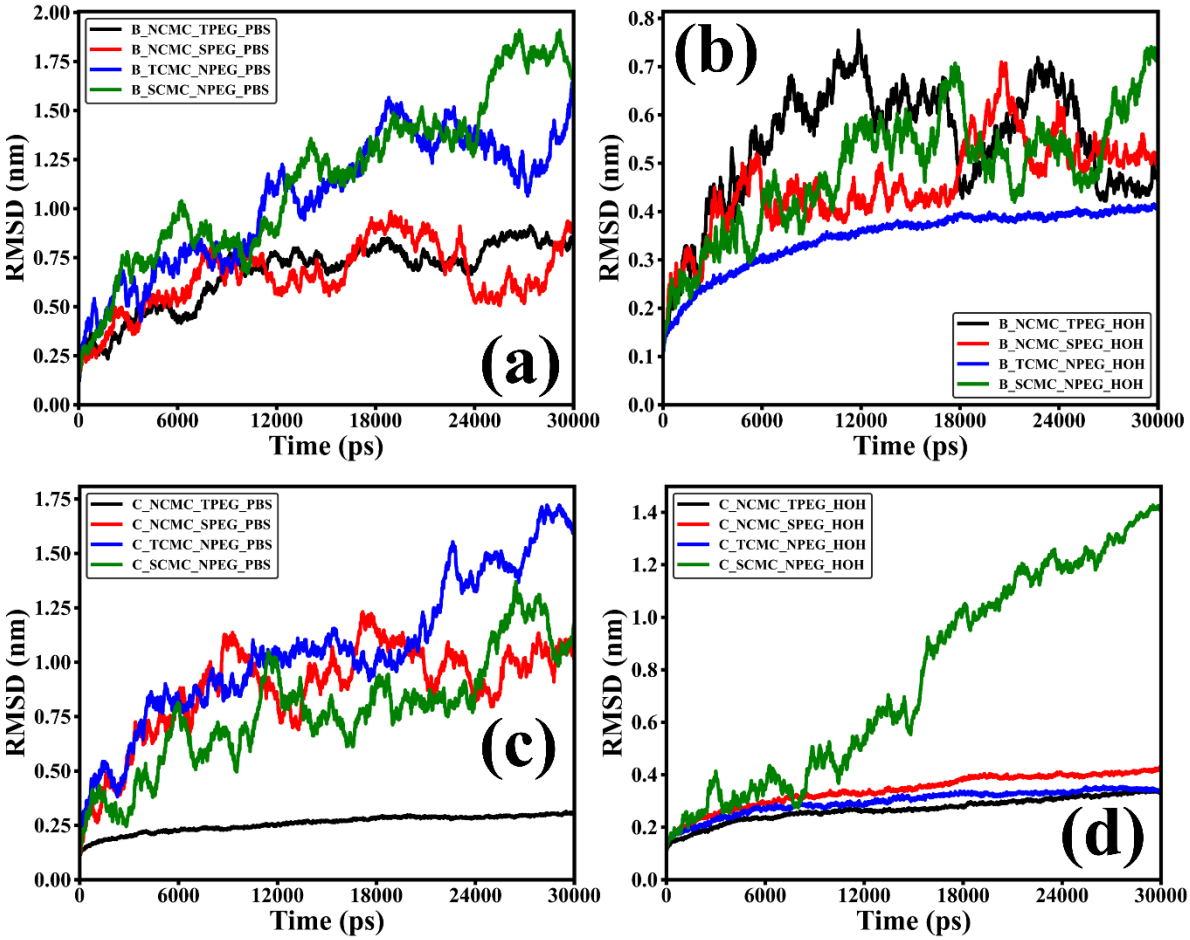


Figure S3: RMSD of polymer framework to polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

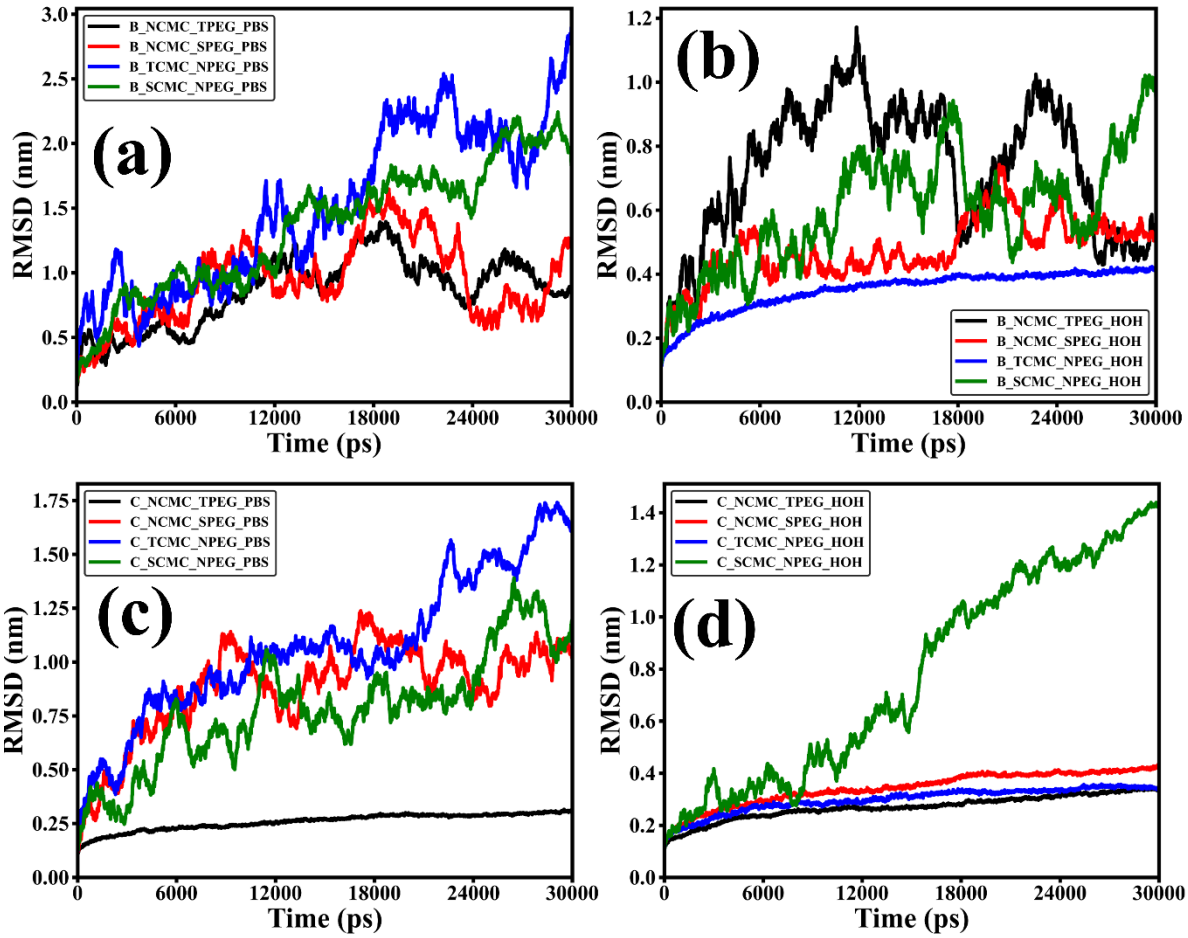


Figure S4. RMSD of CMC based polymer framework to polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

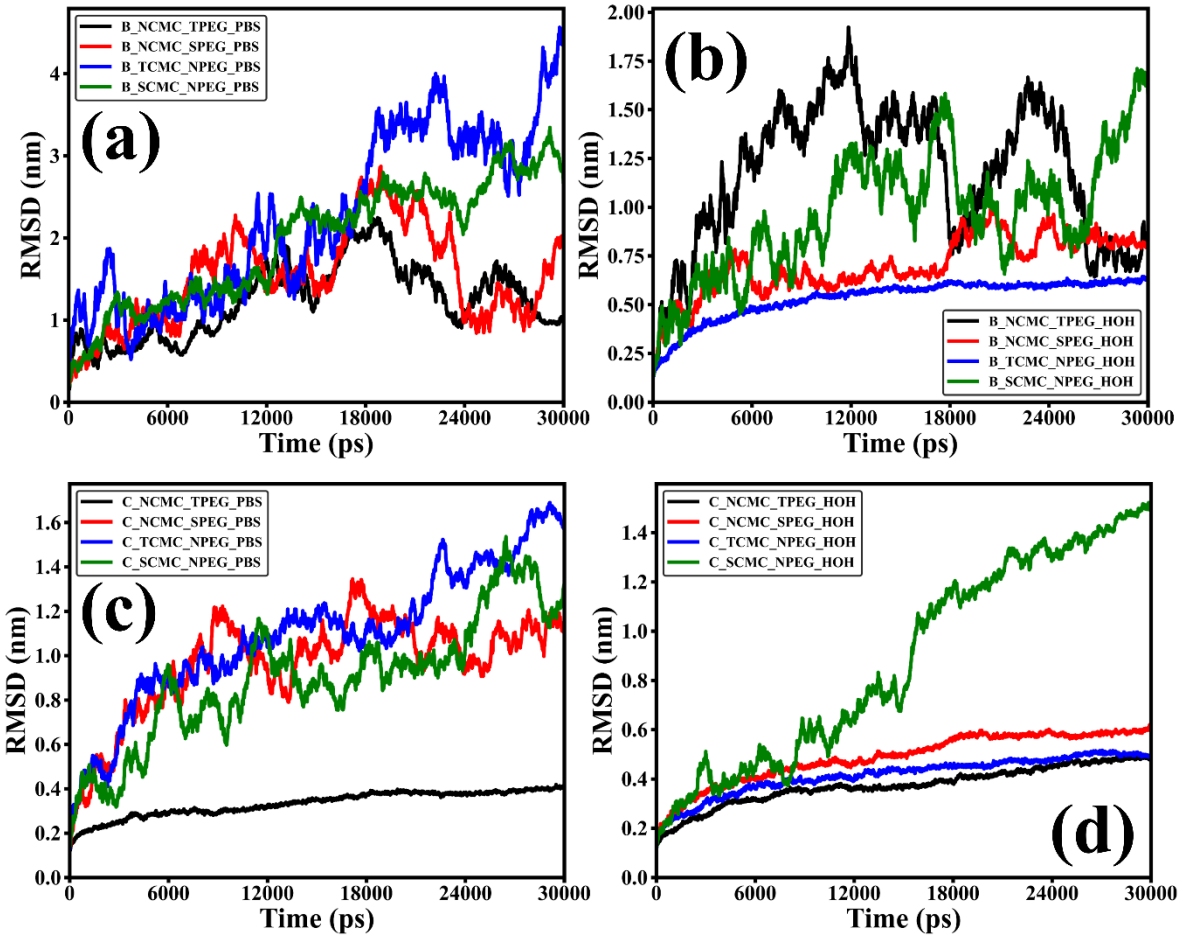


Figure S5. RMSD of PEG based polymer framework to polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

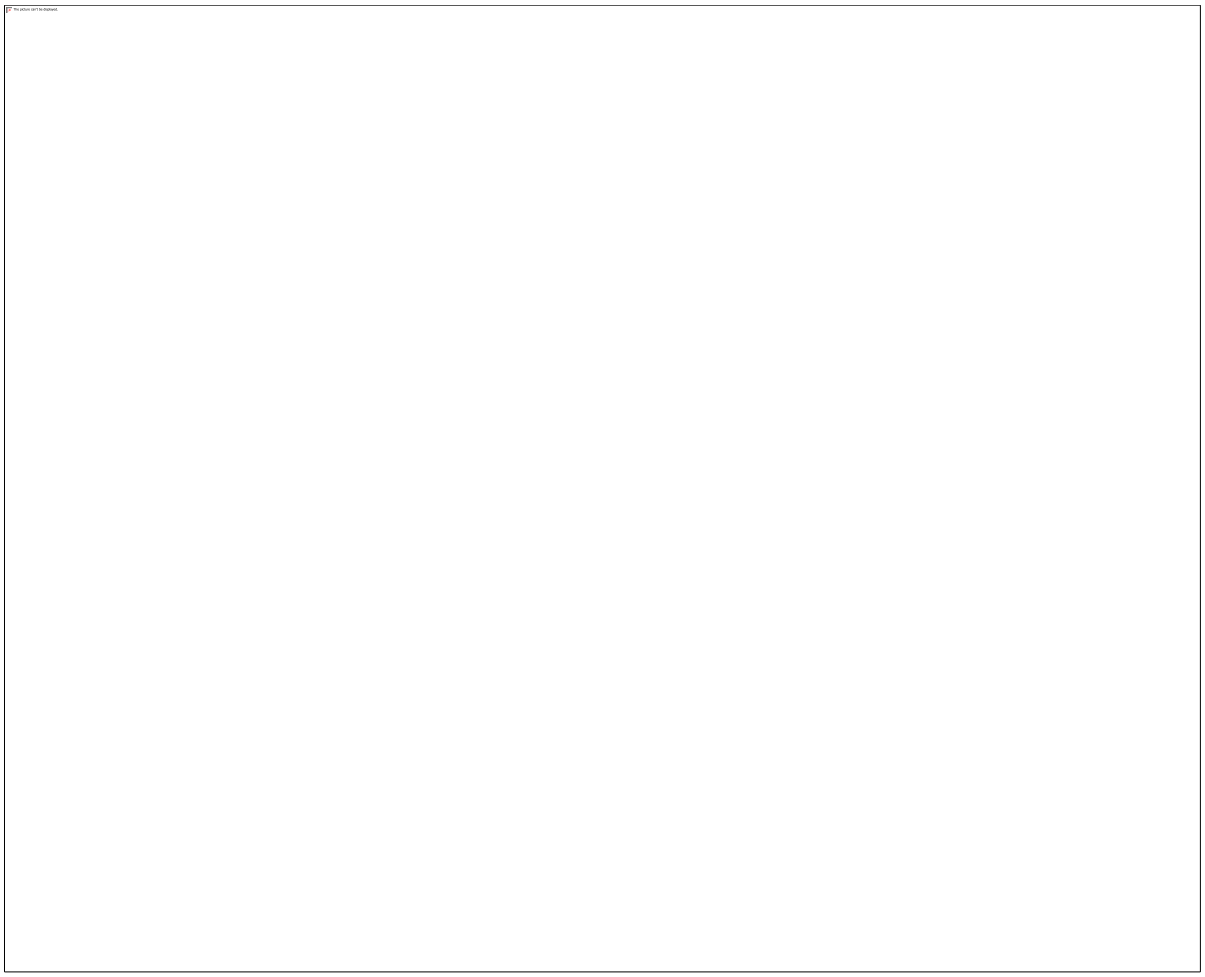


Figure S6. RMSD of CMC based polymer framework to PEG based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

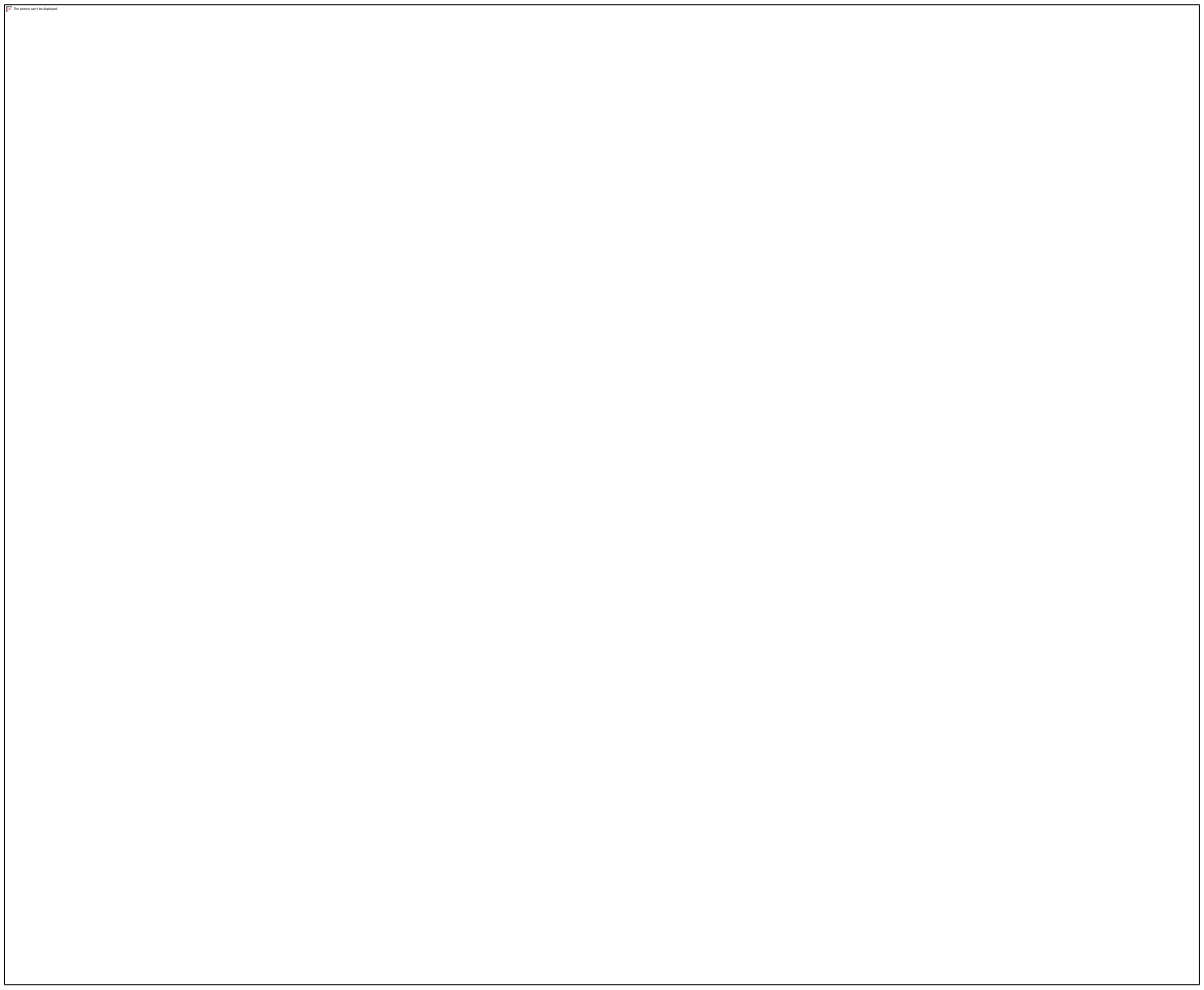


Figure S7. H-bond of solvent molecules to the CMC based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

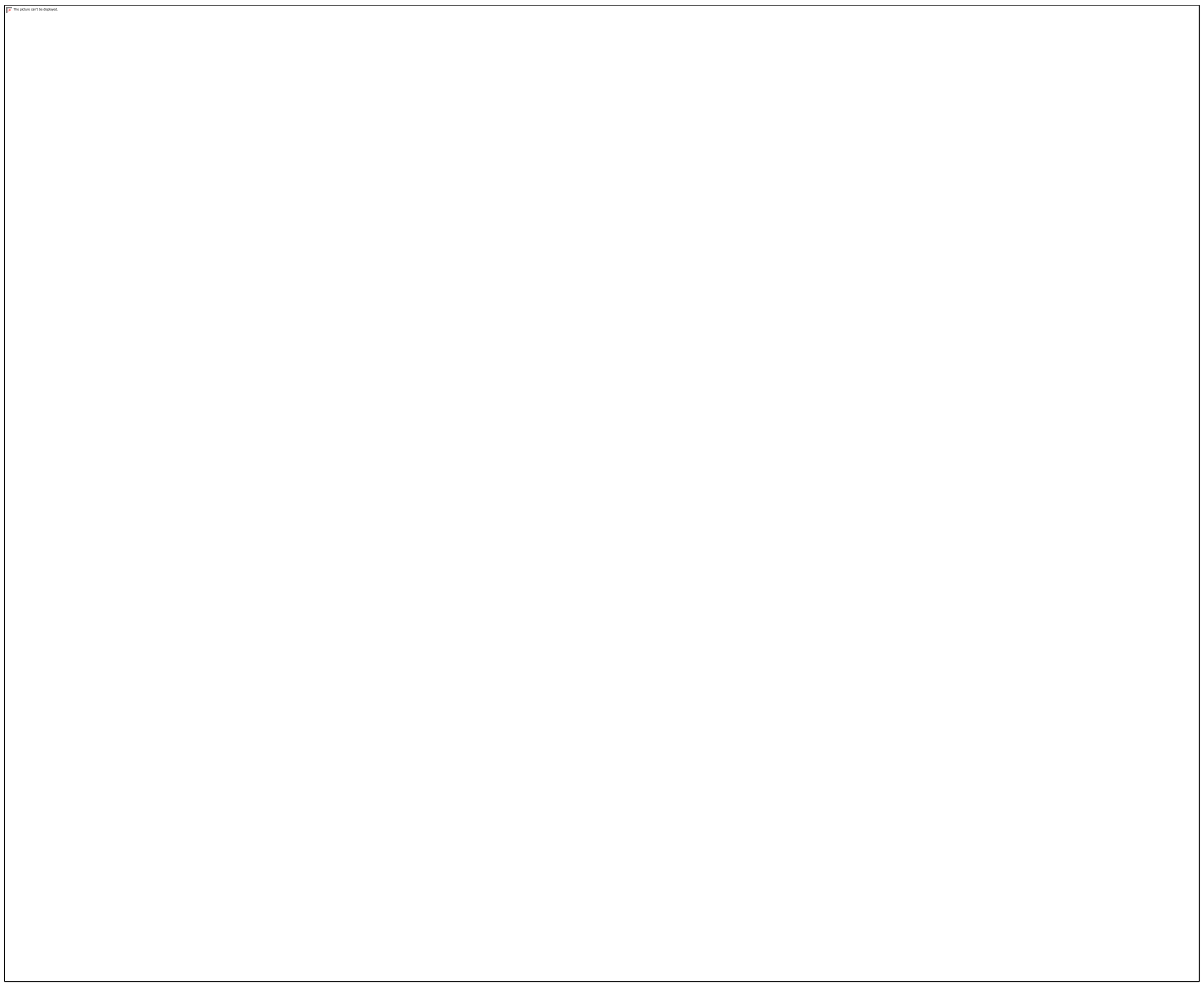


Figure S8. H-bond of solvents molecules to the PEG based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

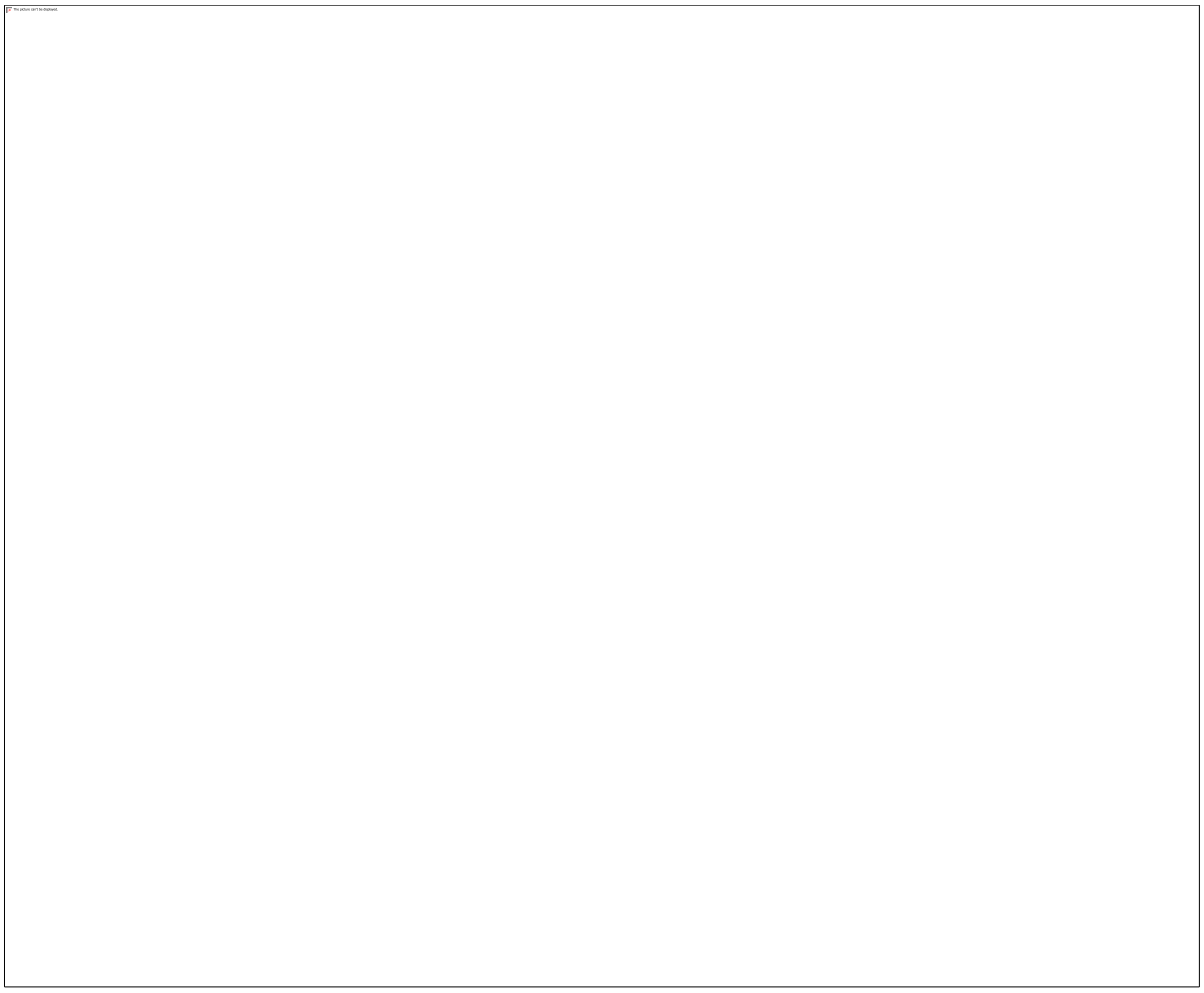


Figure S9. H-bond of PEG based polymer framework to the CMC based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

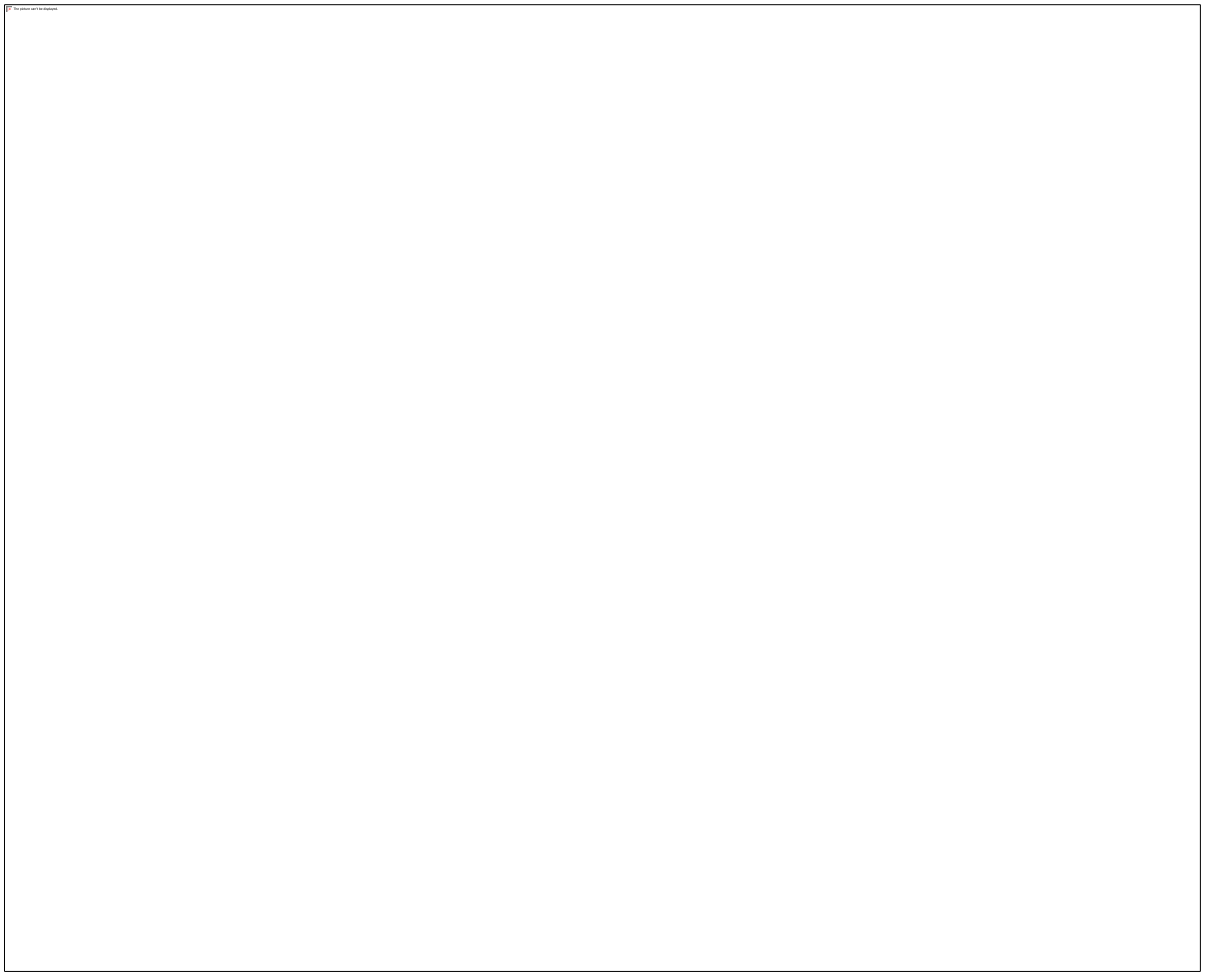


Figure S10. H-bond of drug molecules to the solvent molecules (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

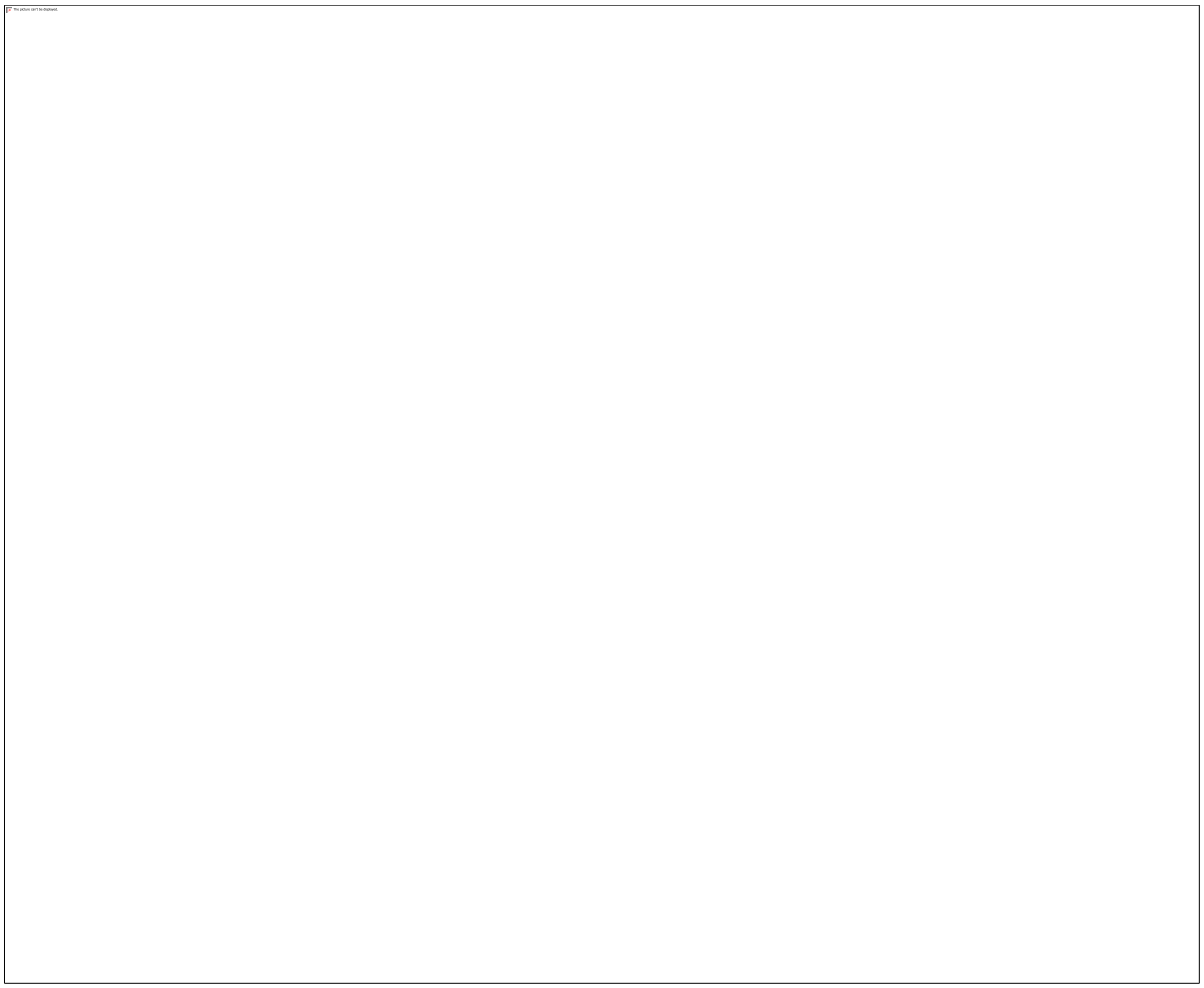


Figure S11. H-bond of drug molecules to the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

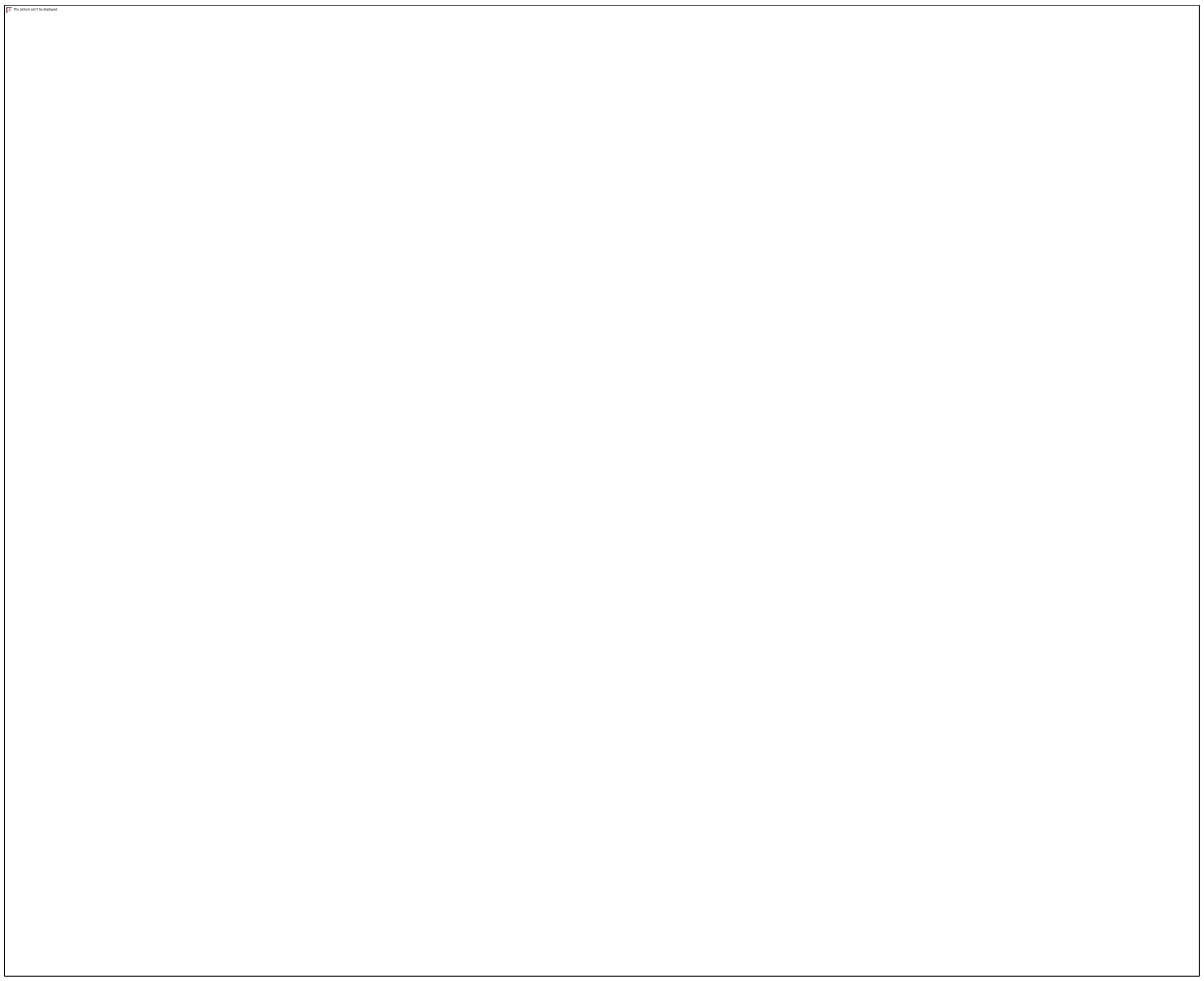


Figure S12. H-bond of drug molecules to the CMC based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

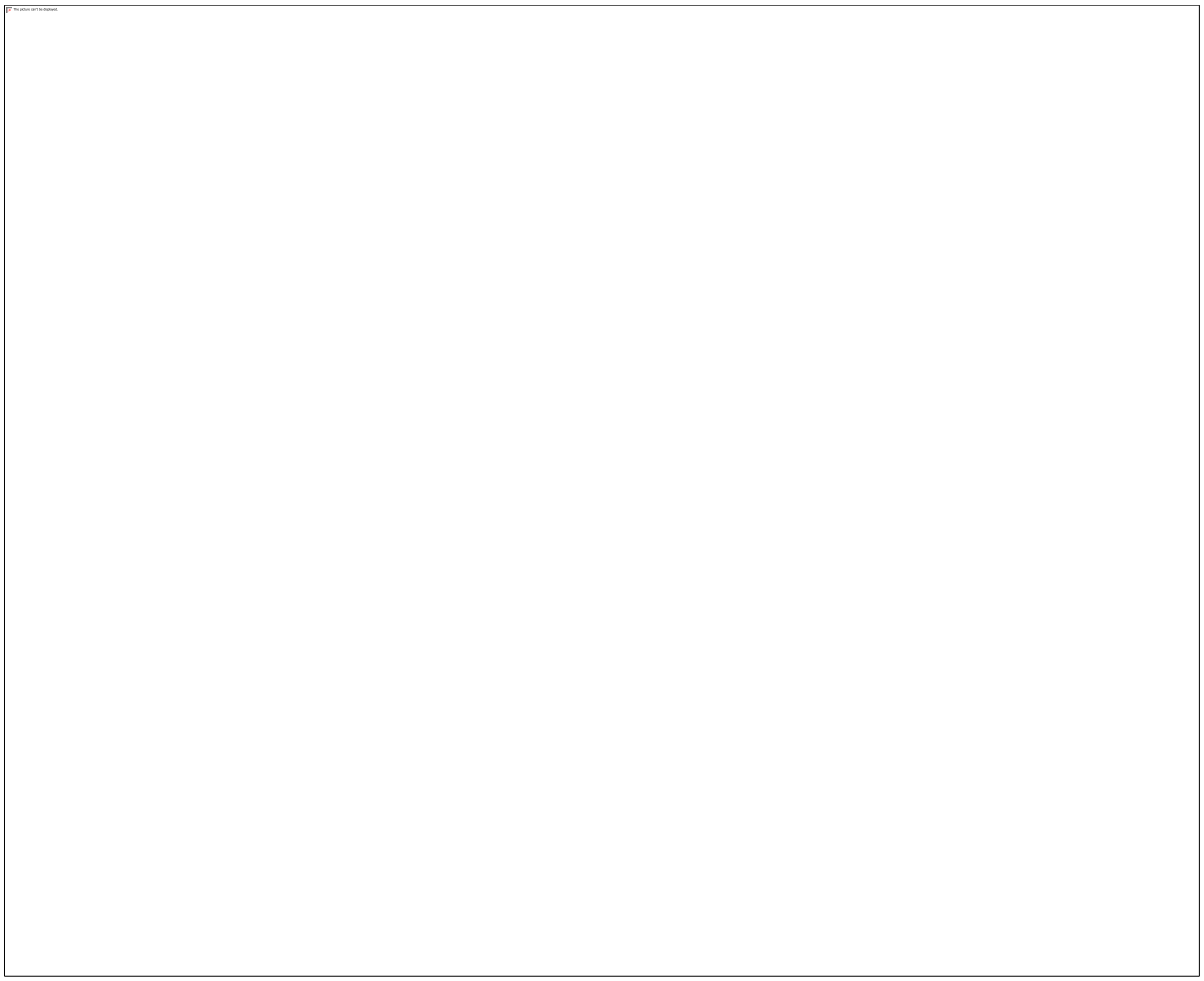


Figure S13. H-bond of drug molecules to the PEG based polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous

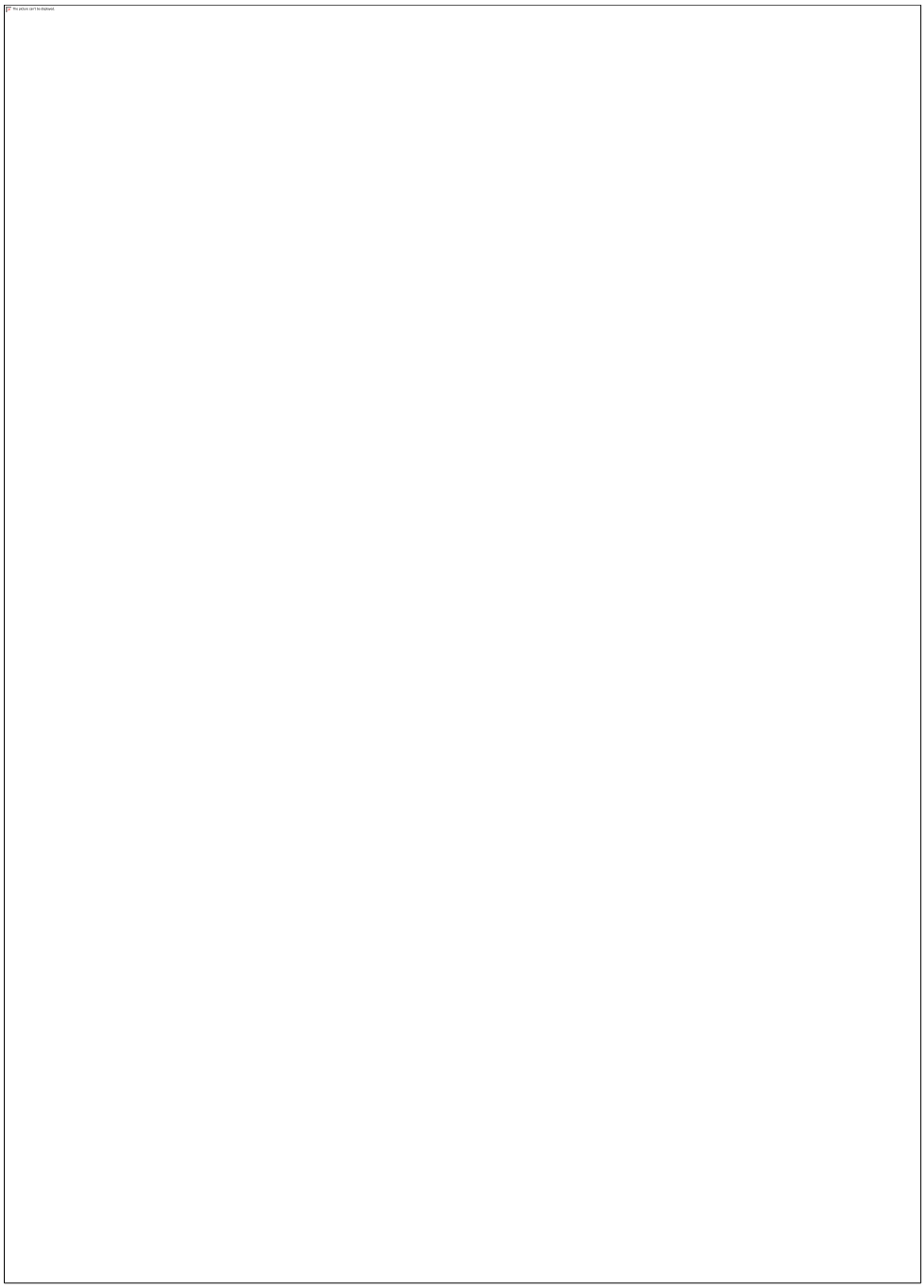


Figure S14. RDF of nitrogen moiety present in the amine group of the ciprofloxacin to the unreacted ene group present in the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of ciprofloxacin and 4-arm-PEG-NB (II) schematic representation of ciprofloxacin and CMC-NB

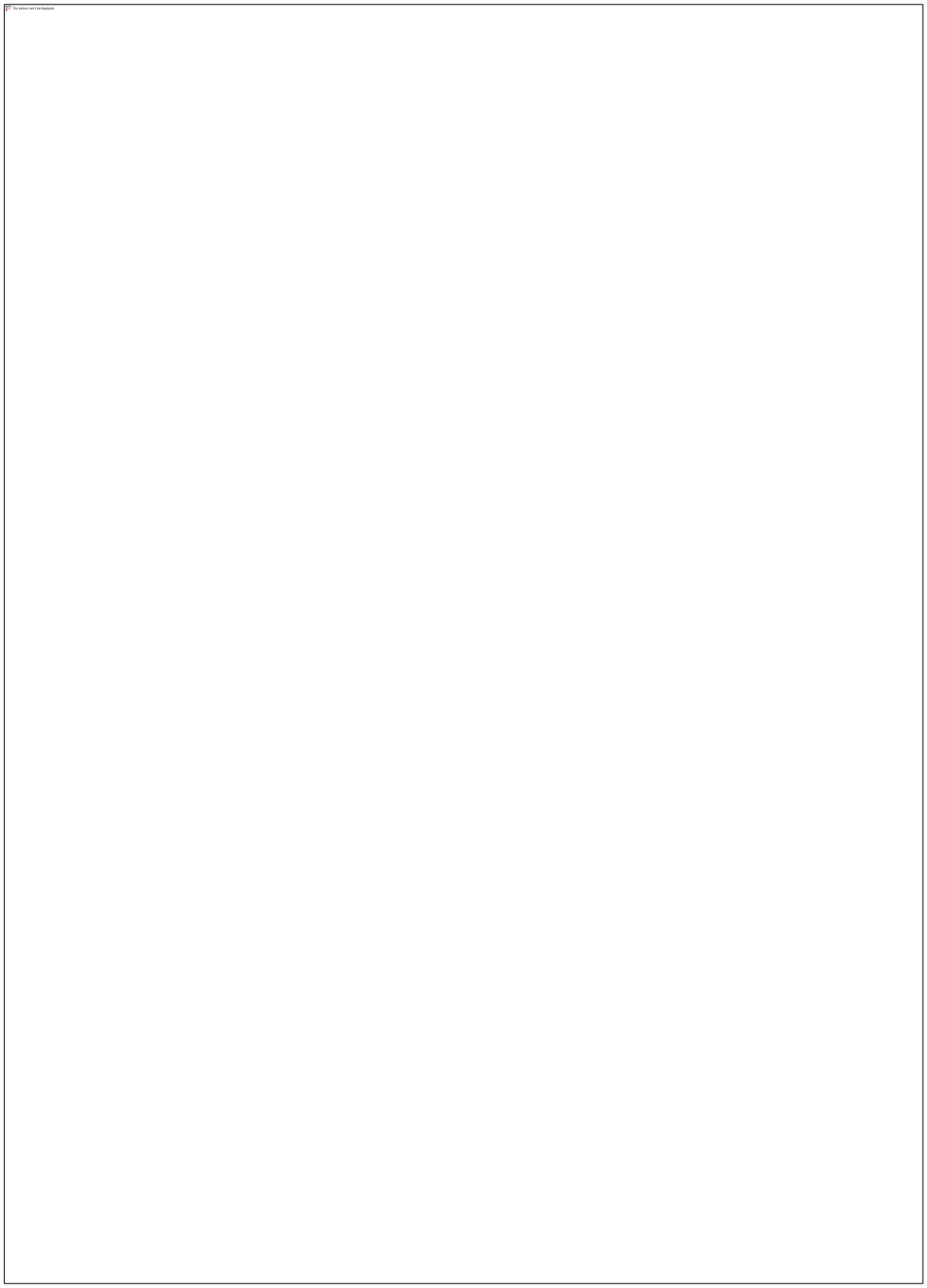


Figure S15. RDF of Flurine moiety present ciprofloxacin to the unreacted ene group present in the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of ciprofloxacin and 4-arm-PEG-NB (II) schematic representation of ciprofloxacin and CMC-

NB

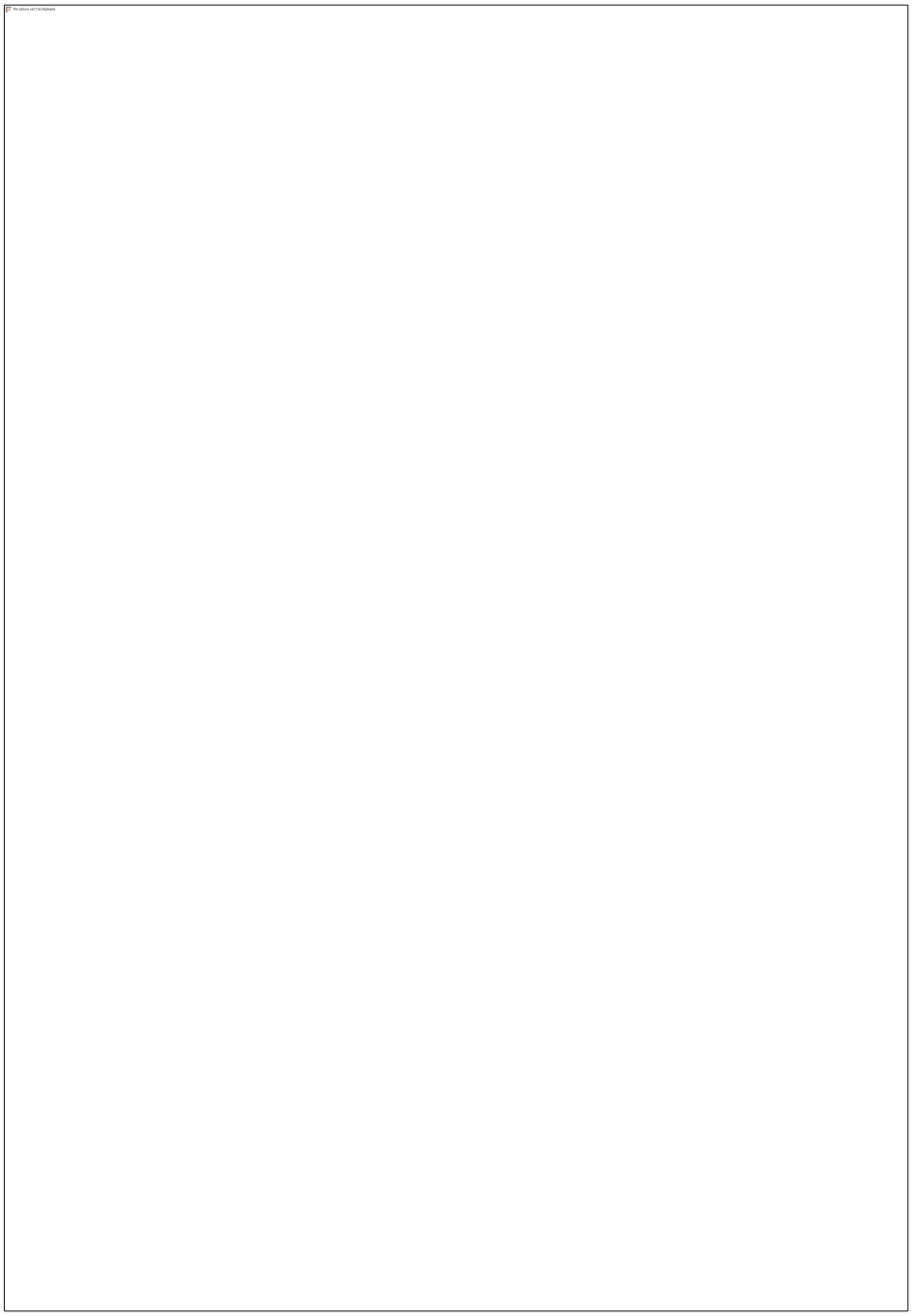


Figure S16. RDF of nitrogen moiety present in the ciprofloxacin to the unreacted ene group present in the polymer framework (a) blend composite in PBS (b) blend composite in aqueous

(c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of ciprofloxacin and 4-arm-PEG-NB (II) schematic representation of ciprofloxacin and CMC-NB

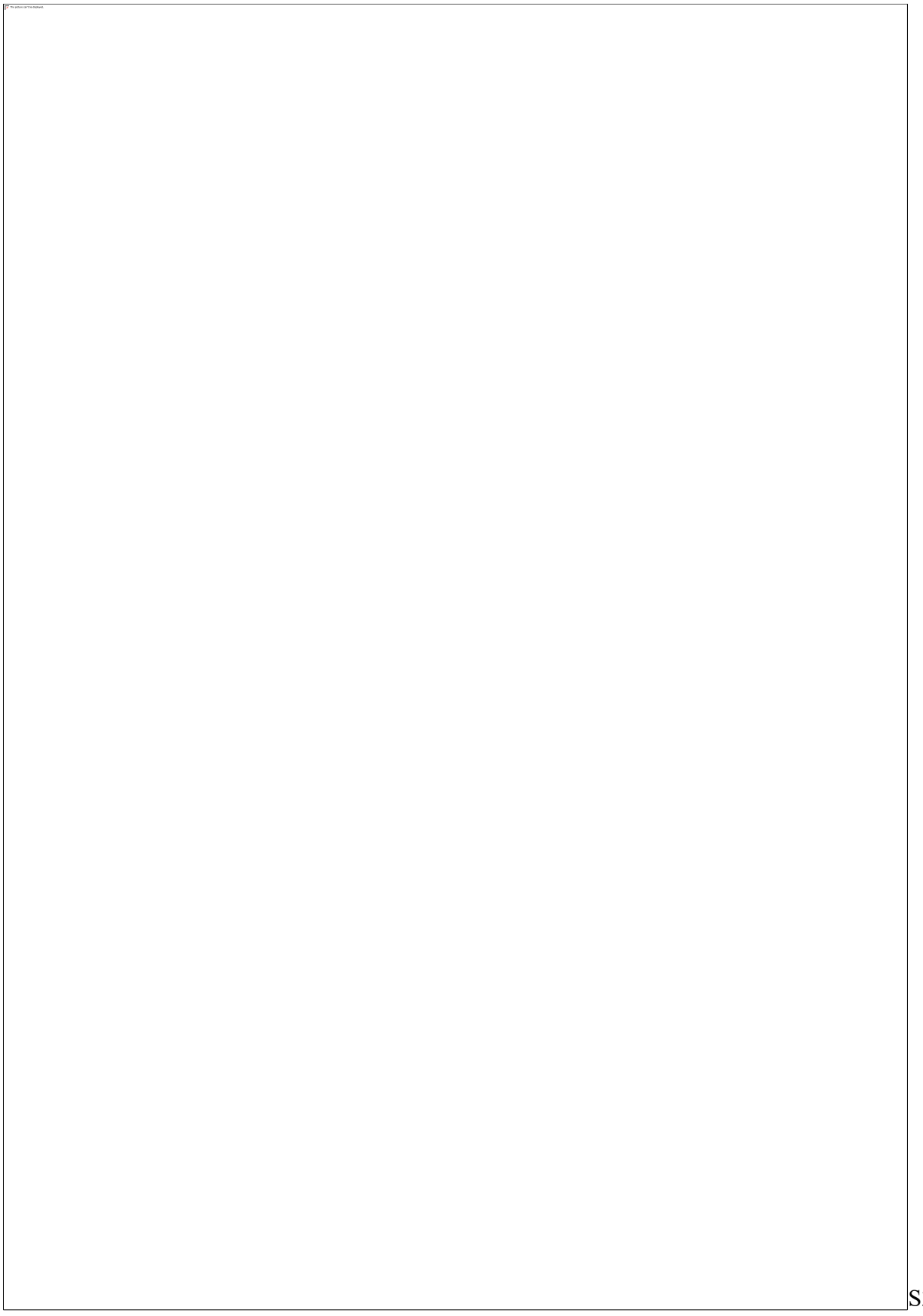




Figure S17. RDF of hydrogen moiety of hydroxyl of carboxylic acid present in the CMC in the polymer framework to the oxygen moiety of ether in the PEG (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of CMC-NB to the PEG-SH (II) schematic representation of CMC-SH to the PEG-NB (III) schematic representation of CMC-NB to the PEG-NB-DTT (IV) schematic representation of CMC-NB-DTT to the PEG-NB

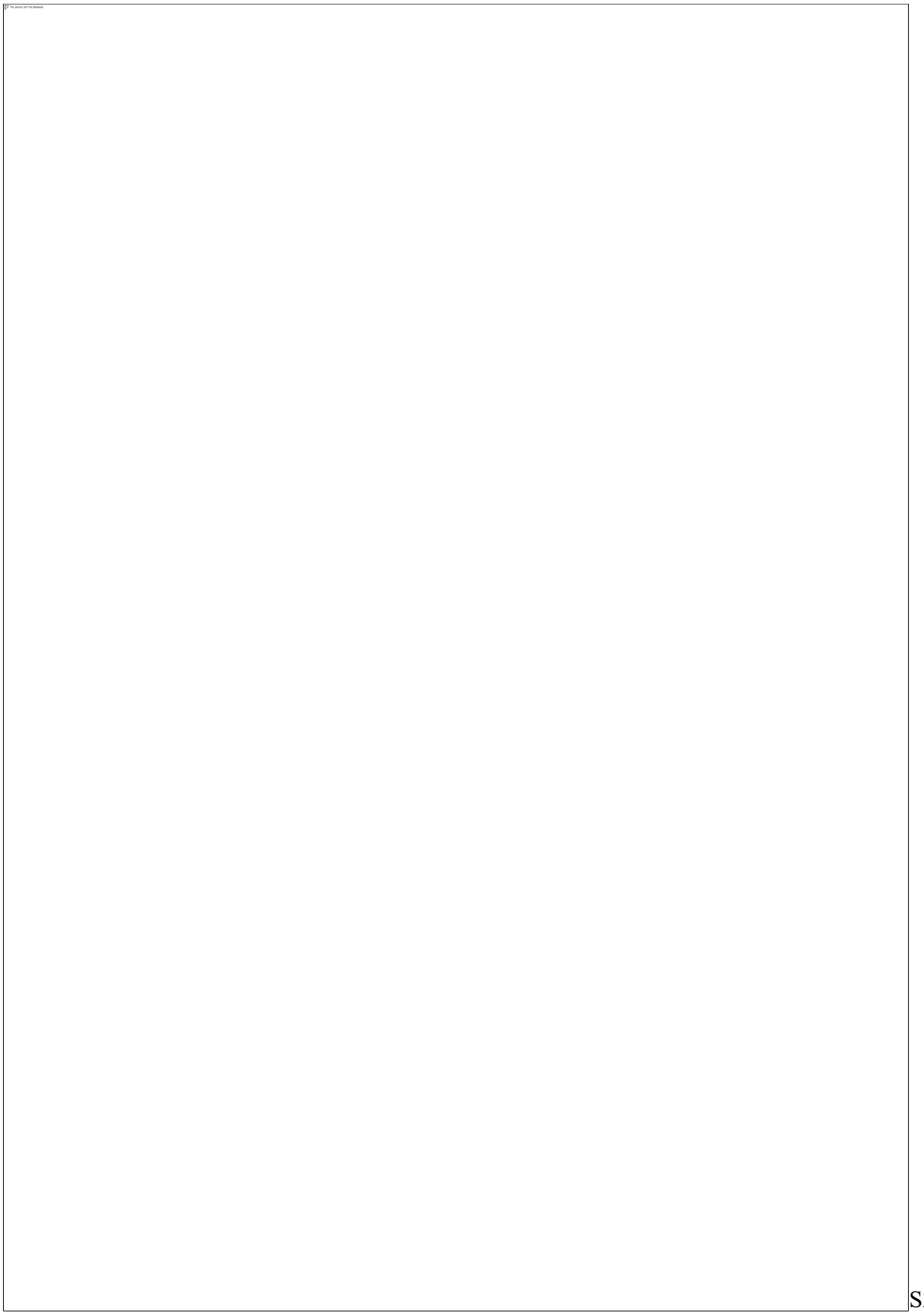
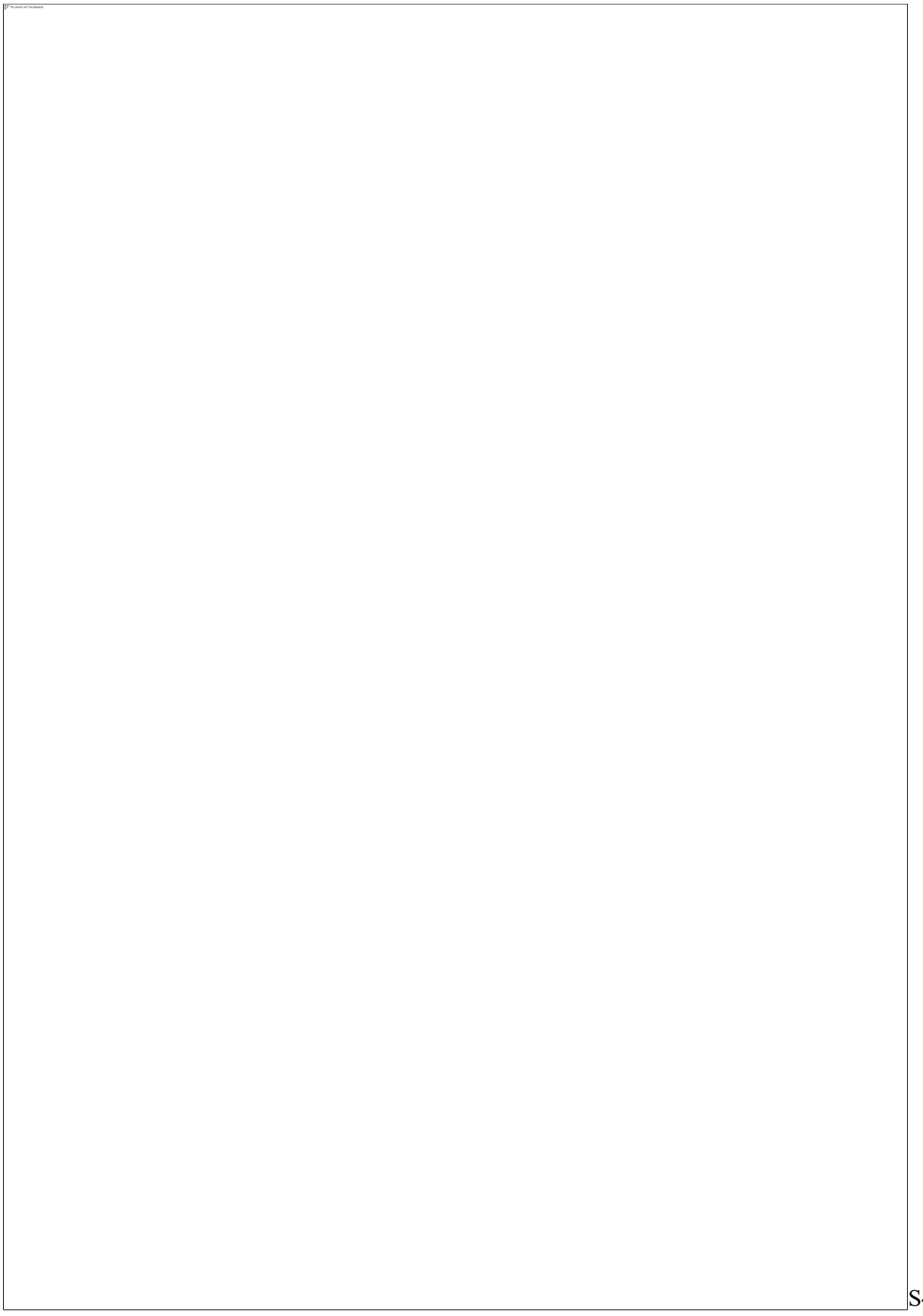




Figure S18. RDF of hydrogen moiety of hydroxyl of the CMC in the polymer framework to the oxygen moiety of ether in the PEG (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of CMC-NB to the PEG-NB-DTT (II) schematic representation of CMC-NB-DTT to the PEG-NB (III) schematic representation of CMC-NB to the PEG-SH (IV) schematic representation of CMC-SH to the PEG-NB



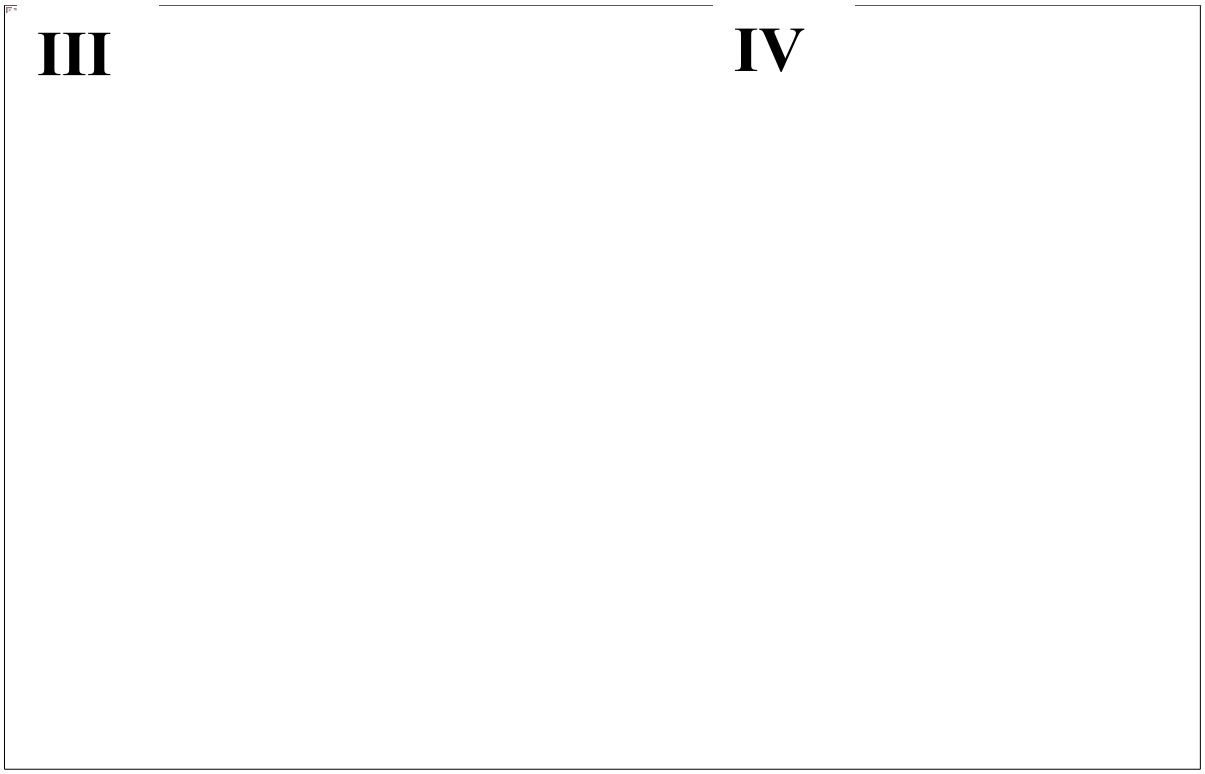


Figure S19. RDF of unreacted ene moiety of the polymer framework to the unreacted thiol moiety of the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of CMC-SH to the PEG-NB (II) schematic representation of CMC-NB to the CMC-NB-DTT (III) schematic representation of PEG-NB-DTT to the PEG-NB (IV) schematic representation of CMC-NB to the PEG-SH

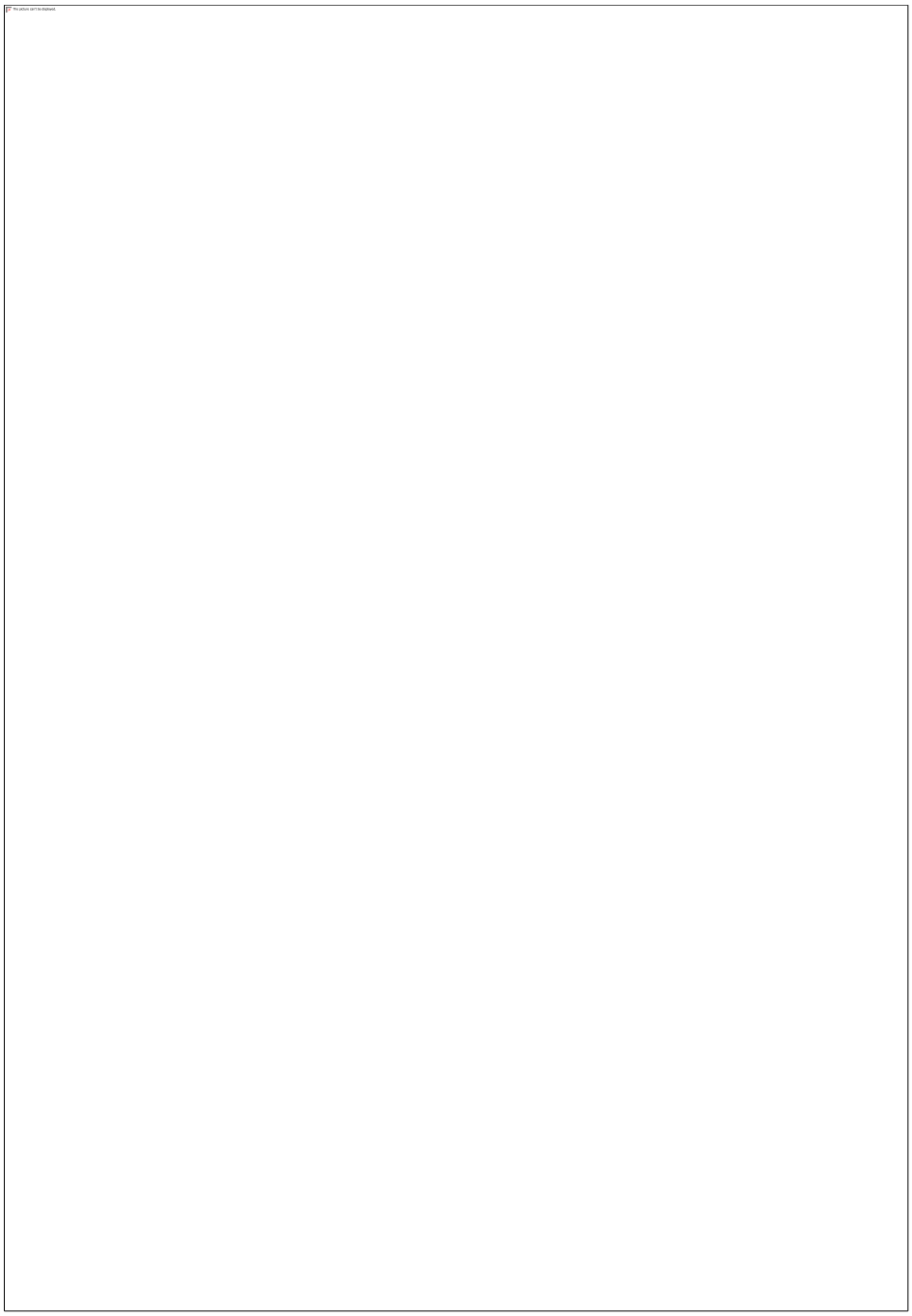


Figure S20. MSD of unreacted ene group present in the blend system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the PBS media

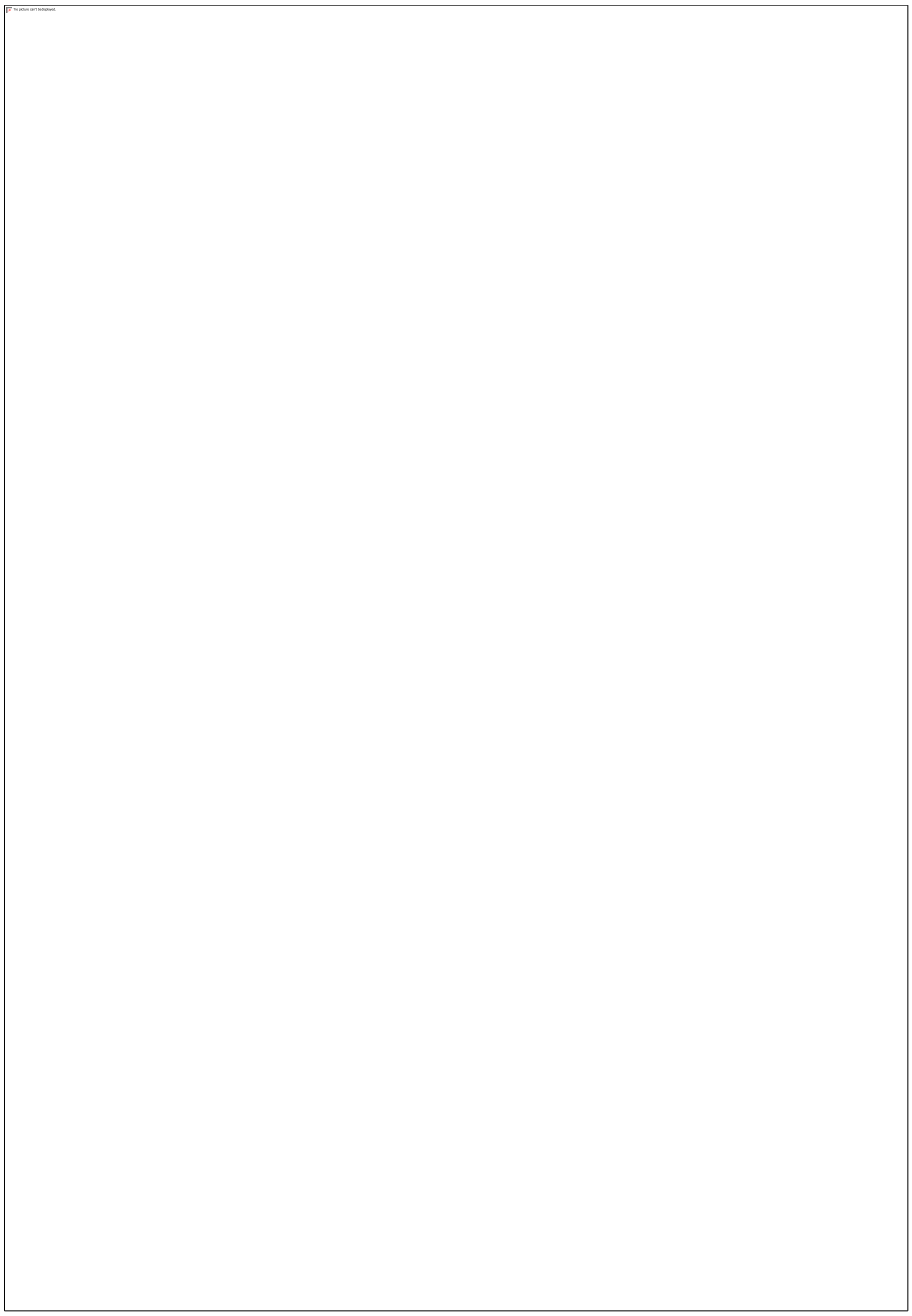


Figure S21. MSD of unreacted thiol group present in the blend system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the PBS media

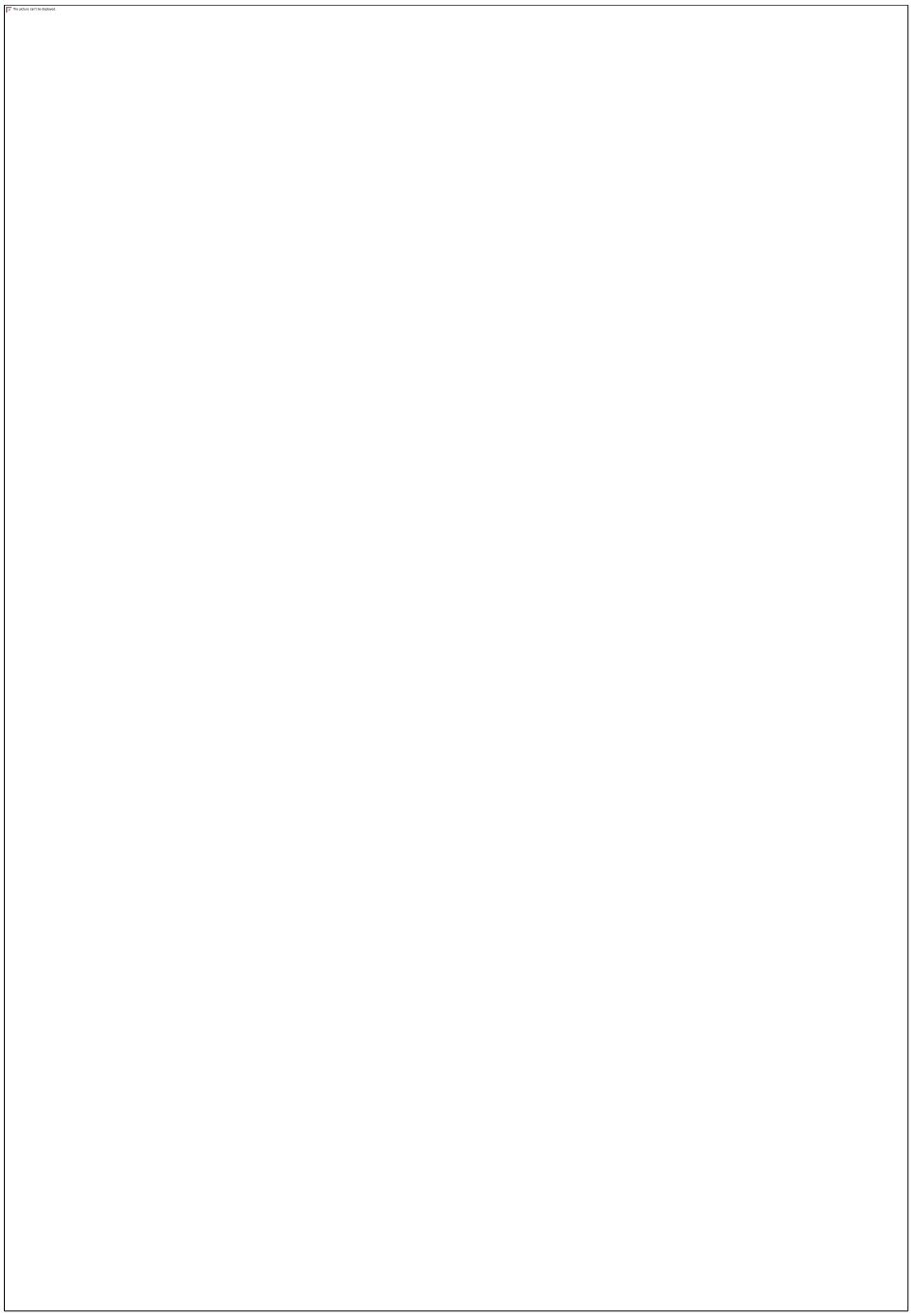


Figure S22. MSD of unreacted ene group present in the copolymer system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the PBS media

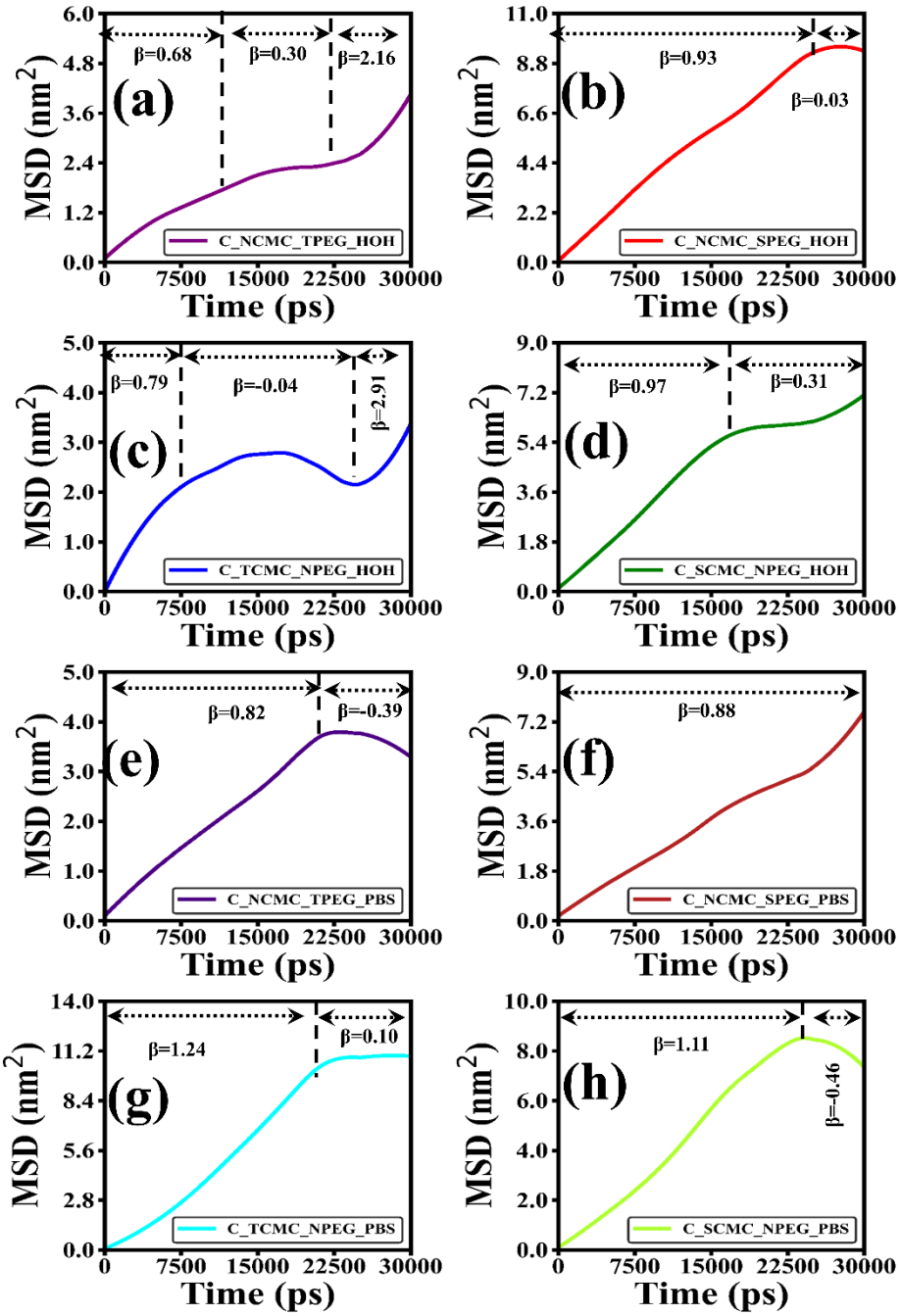


Figure S23. MSD of unreacted thiol group present in the copolymer system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the

PBS media